

Chapter 4: The Seaborn essentials for data visualization

Did you use an LLM?

```
In [ ]: ##### Yes I used LLM but only for comments and only mundane ones but other than t
```

```
In [1]: #Import Matplotlib library for plotting
import matplotlib as plt
```

```
In [2]: # Import Pandas library for data manipulation and analysis
import pandas as pd
```

Get the data

```
In [3]: # Load the mortality data from a preprocessed pickle file into a Pandas DataFrame
mortality_data = pd.read_pickle('mortality_prepped.pkl')
mortality_data.head()
```

```
Out[3]:
```

	Year	AgeGroup	DeathRate	MeanCentered
0	1900	01-04 Years	1983.8	1790.87584
1	1901	01-04 Years	1695.0	1502.07584
2	1902	01-04 Years	1655.7	1462.77584
3	1903	01-04 Years	1542.1	1349.17584
4	1904	01-04 Years	1591.5	1398.57584

The basic Seaborn parameters

```
In [4]: # Import Seaborn library for data visualization
import seaborn as sns
```

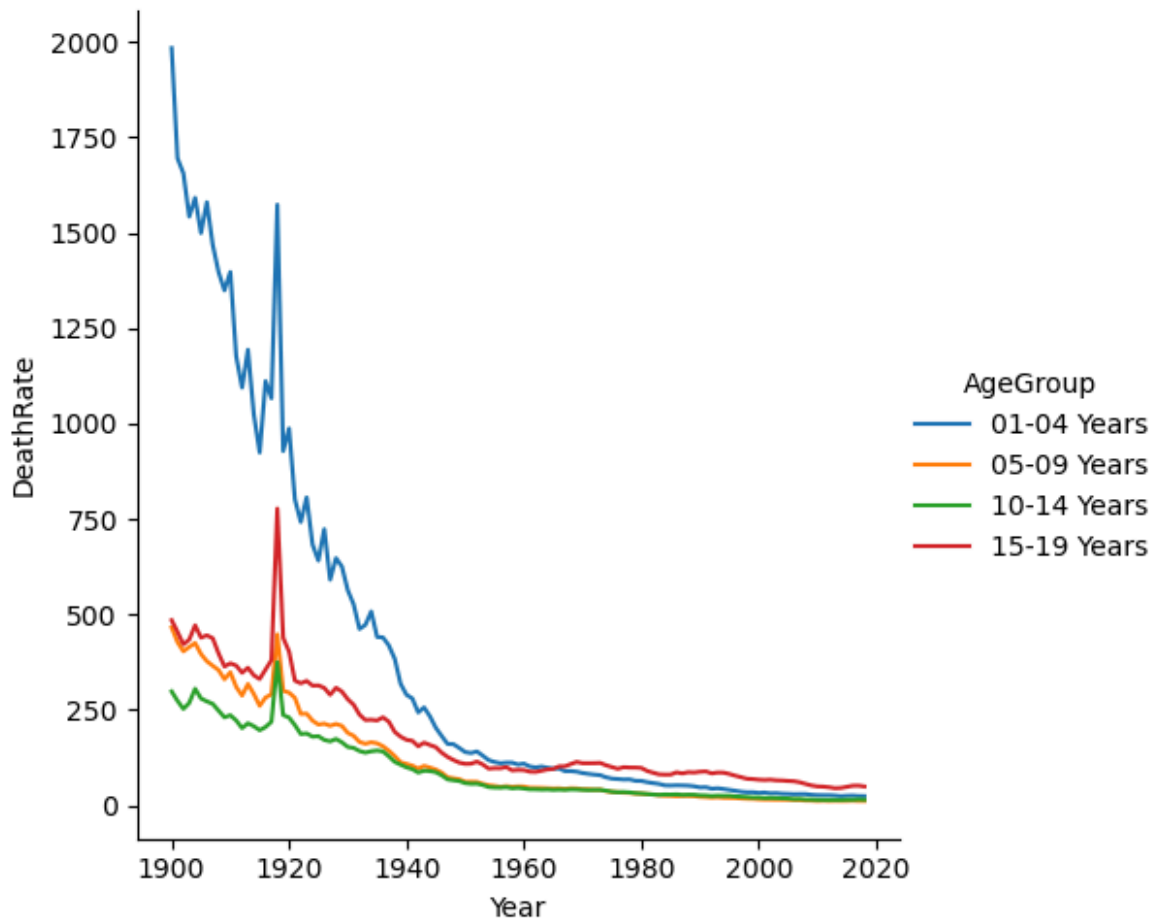
```
In [5]: # Created a line plot using Seaborn to visualize the mortality data with X axis
sns.relplot(data=mortality_data, kind='line',
            x='Year', y='DeathRate', hue='AgeGroup')
```

```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
rt inf values to NaN before operating instead.
    with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
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rt inf values to NaN before operating instead.
    with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: Th
e figure layout has changed to tight
    self.figure.tight_layout(*args, **kwargs)

```

Out[5]: <seaborn.axisgrid.FacetGrid at 0x2cdc91ca5d0>



```

In [6]: # Create a line plot with Seaborn with X axis as Year, hue as Age Group and Y as
sns.lineplot(data=mortality_data,
             x='Year', y='DeathRate', hue='AgeGroup')

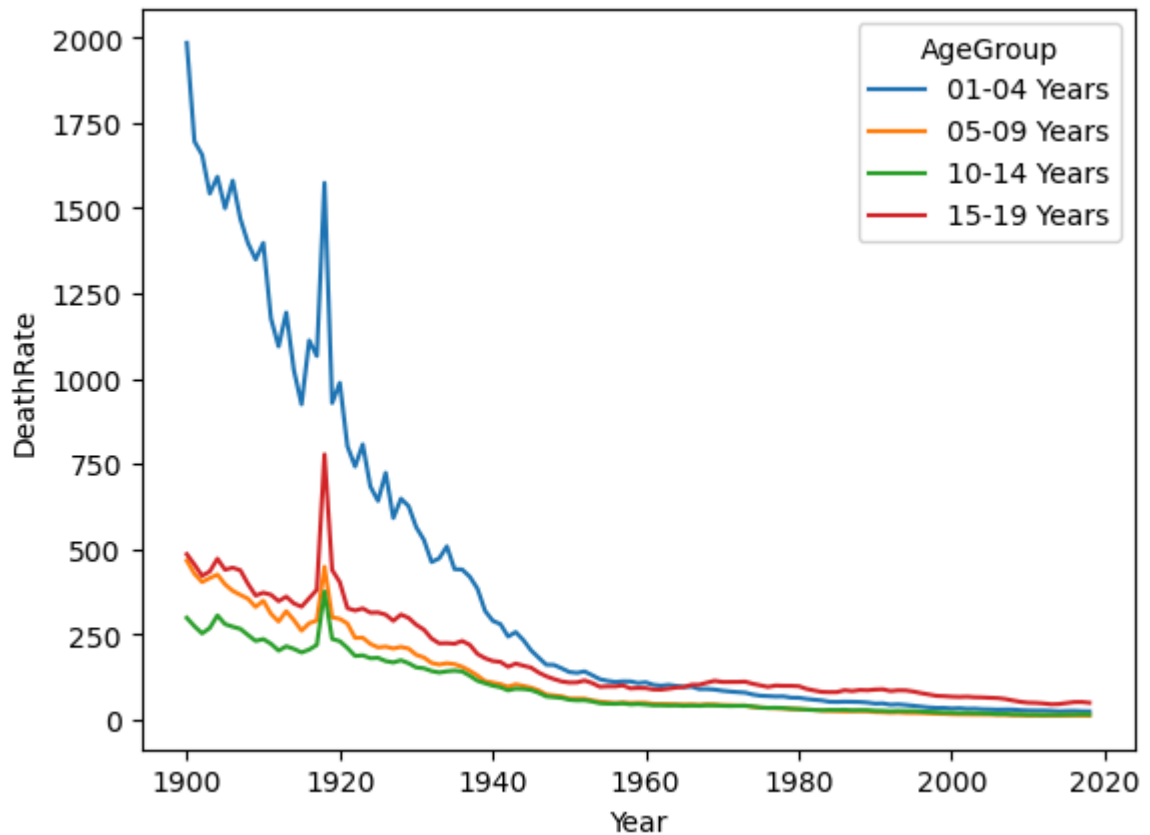
```

```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
rt inf values to NaN before operating instead.
    with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
rt inf values to NaN before operating instead.
    with pd.option_context('mode.use_inf_as_na', True):

```

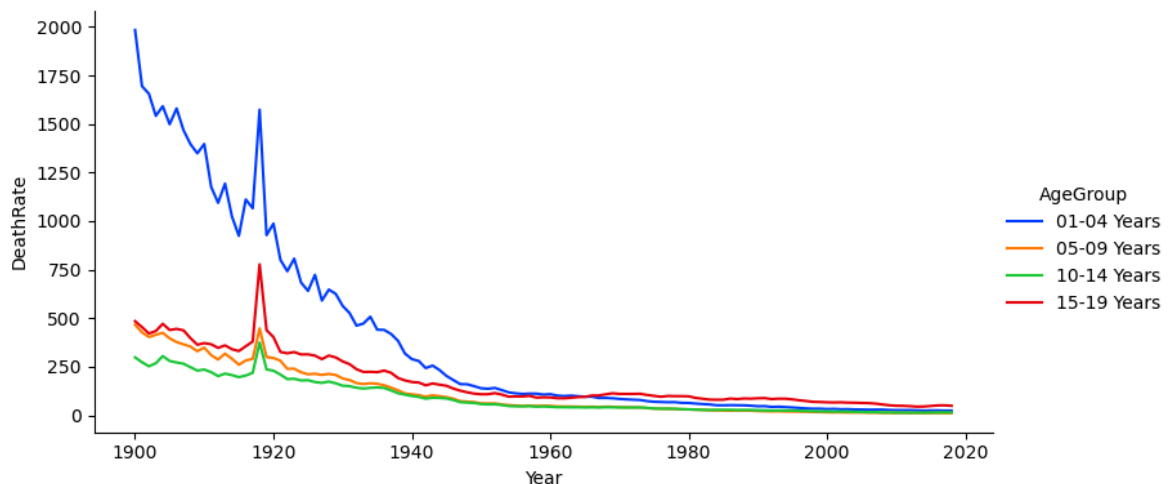
Out[6]: <Axes: xlabel='Year', ylabel='DeathRate'>

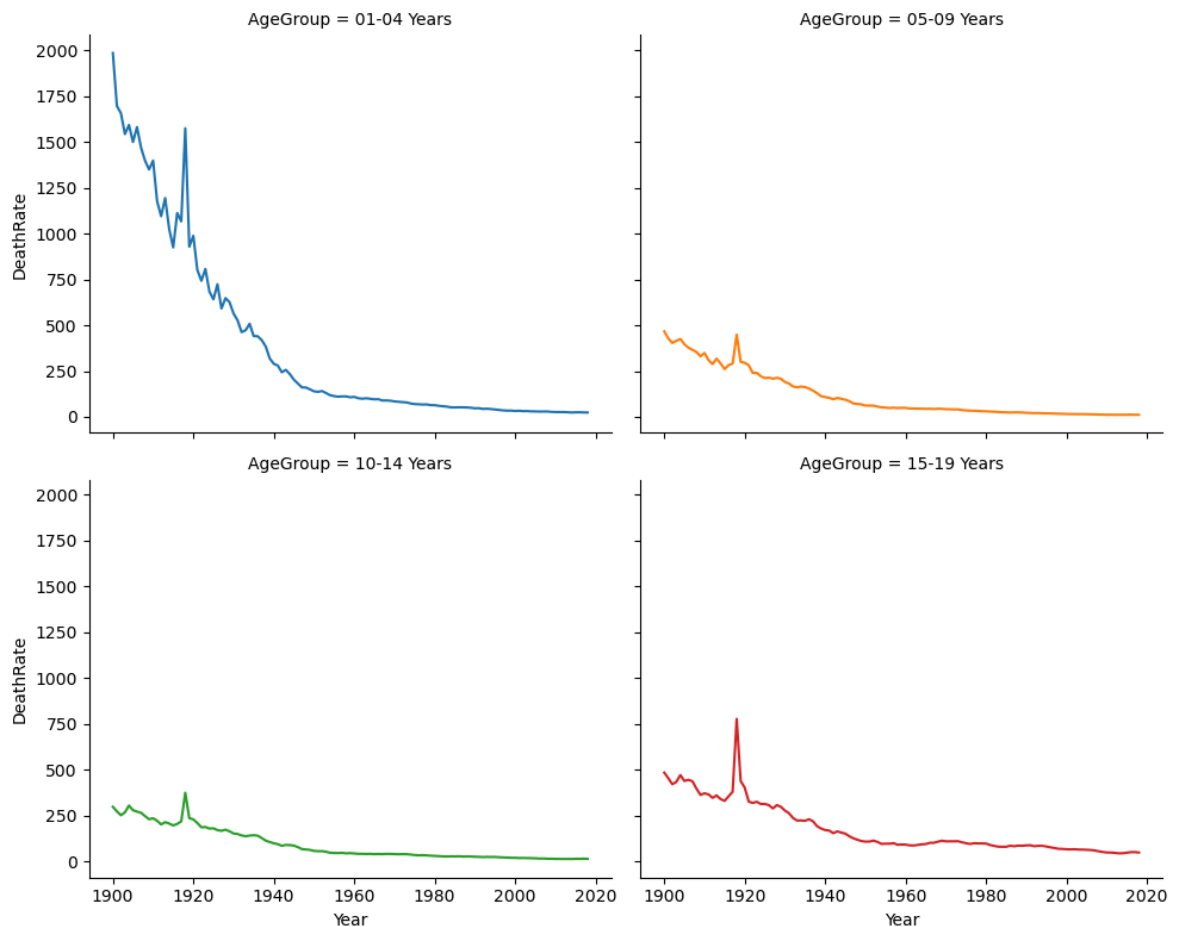


```
In [8]: # Another variation of a line plot with Seaborn with aspect changed to 2 from 1.
sns.relplot(data=mortality_data, kind='line',
            x='Year', y='DeathRate', hue='AgeGroup', palette='bright',
            height=4, aspect=2)
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.
 with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.
 with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The figure layout has changed to tight
 self._figure.tight_layout(*args, **kwargs)

Out[8]: <seaborn.axisgrid.FacetGrid at 0x2cdc9d6a5d0>





Enhancing plots

```
In [9]: # Customized a Line plot with Seaborn, including setting the axis labels and plot
ax = sns.lineplot(data=mortality_data,
                  x='Year', y='DeathRate', hue='AgeGroup')
ax.set(title='Deaths by Age Group', ylabel='Deaths per 100,000')
```

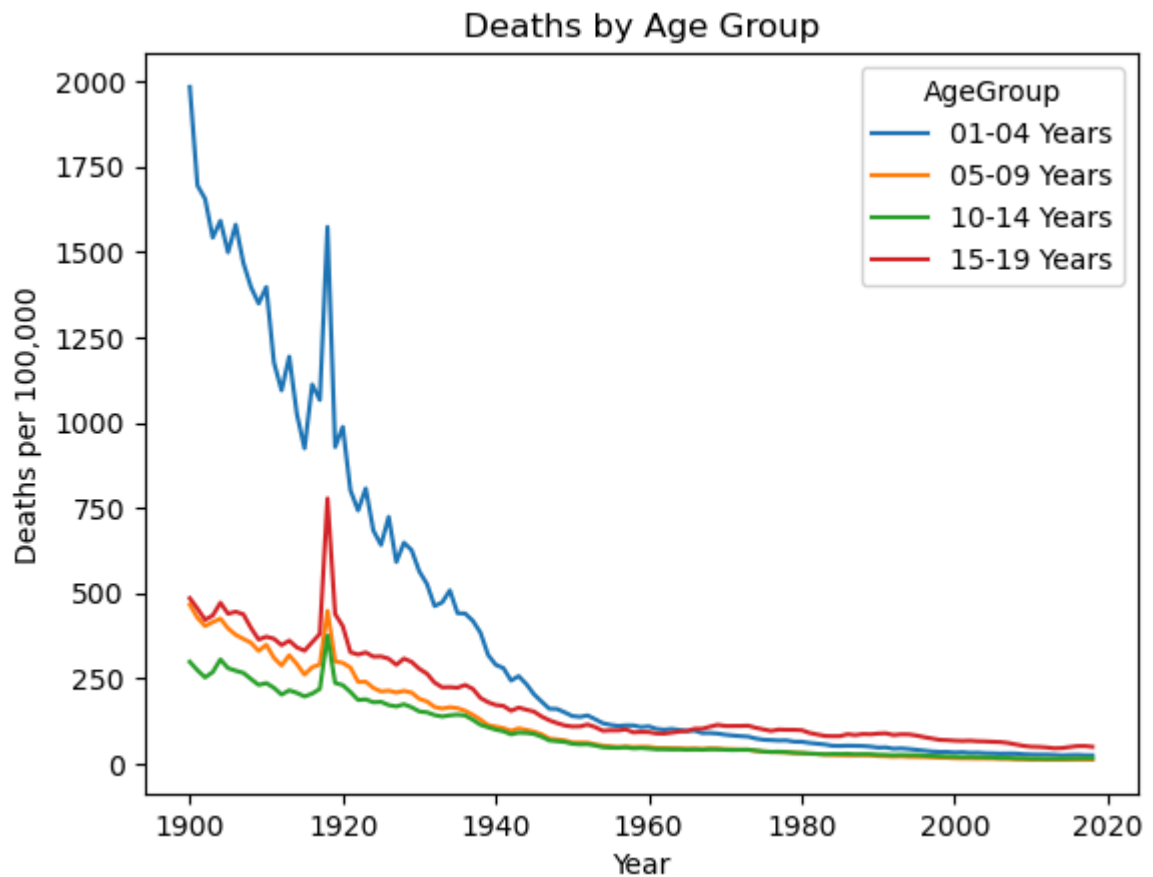
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

with pd.option_context('mode.use_inf_as_na', True):

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

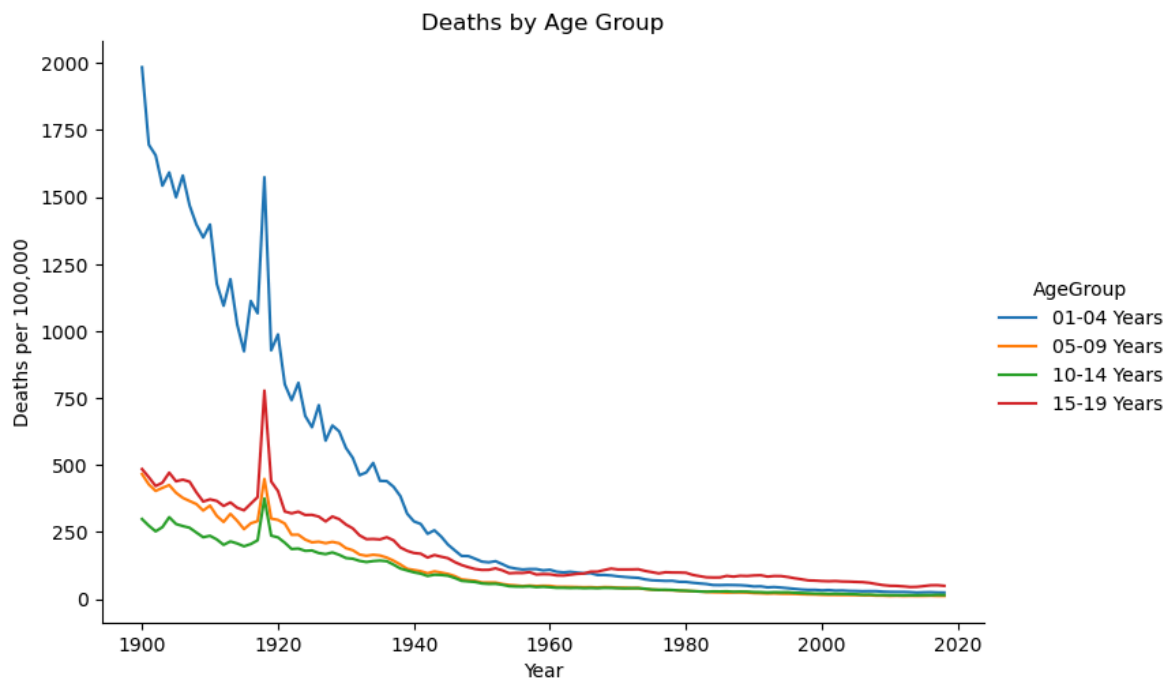
with pd.option_context('mode.use_inf_as_na', True):

```
Out[9]: [Text(0.5, 1.0, 'Deaths by Age Group'), Text(0, 0.5, 'Deaths per 100,000')]
```



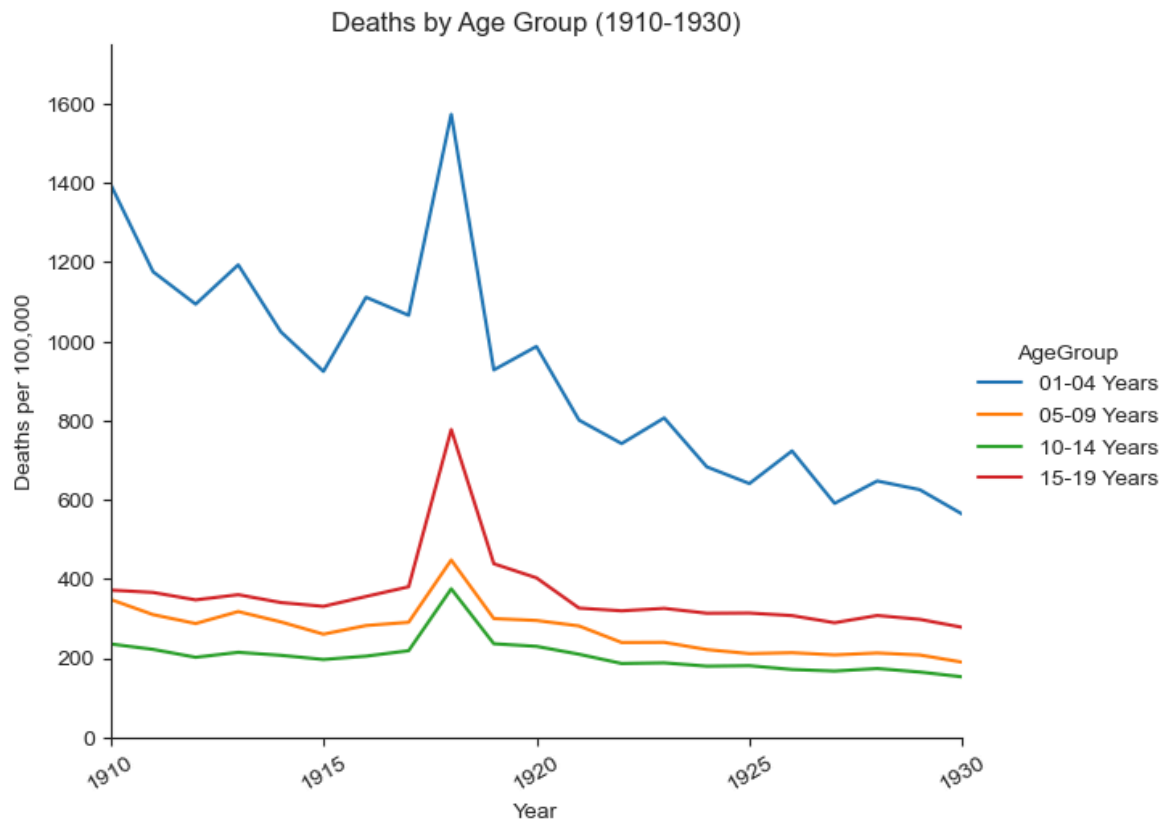
```
In [10]: # Created a relational plot with Seaborn, grouping data by 'AgeGroup' and custom
g = sns.relplot(data=mortality_data, kind='line',
                x='Year', y='DeathRate', hue='AgeGroup', aspect=1.5)
for ax in g.axes.flat:
    ax.set(title='Deaths by Age Group', ylabel='Deaths per 100,000')
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.
 with pd.option_context('mode.use_inf_as_na', True):
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 with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The figure layout has changed to tight
 self._figure.tight_layout(*args, **kwargs)



```
In [53]: # Changed cell 10 so every fifth year is displayed on the x-axis for the plot and
g = sns.relplot(data=mortality_data, kind='line',
                x='Year', y='DeathRate', hue='AgeGroup', aspect=1.25)
for ax in g.axes.flat:
    ax.set(title='Deaths by Age Group (1910-1930)', ylabel='Deaths per 100,000',
           xticks=[x for x in range(1910, 1931, 5)],
           xlim=(1910, 1930), ylim=(0, 1750)),
    ax.tick_params('x', labelrotation=30)
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.
 with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.
 with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The figure layout has changed to tight
 self._figure.tight_layout(*args, **kwargs)



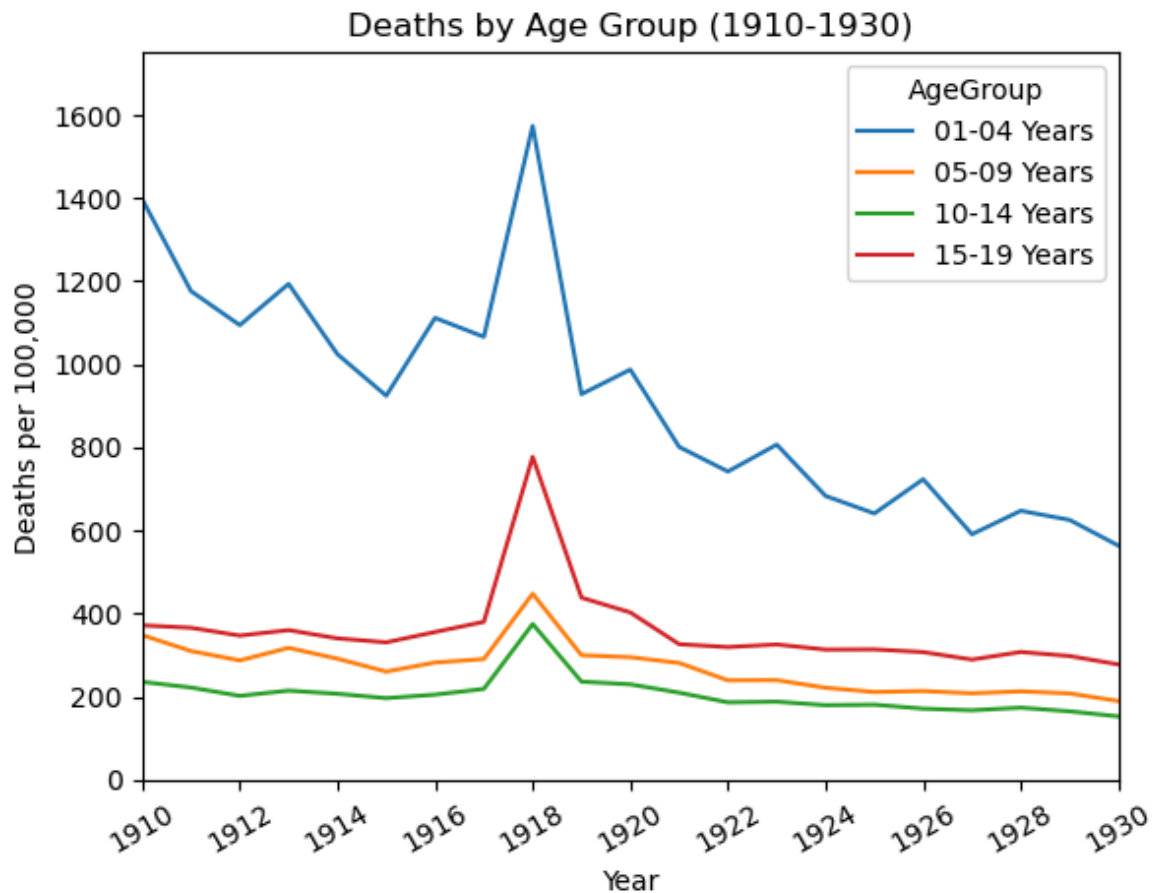
```
In [12]: #Another example of a relational plot with Seaborn, this time focusing on a spec
ax = sns.lineplot(data=mortality_data,
                  x='Year', y='DeathRate', hue='AgeGroup')
ax.set(title='Deaths by Age Group (1910-1930)',
      ylabel='Deaths per 100,000',
      xticks=[x for x in range(1910,1931,2)],
      xlim=(1910,1930), ylim=(0,1750))
ax.tick_params('x', labelrotation=30)
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

with pd.option_context('mode.use_inf_as_na', True):

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

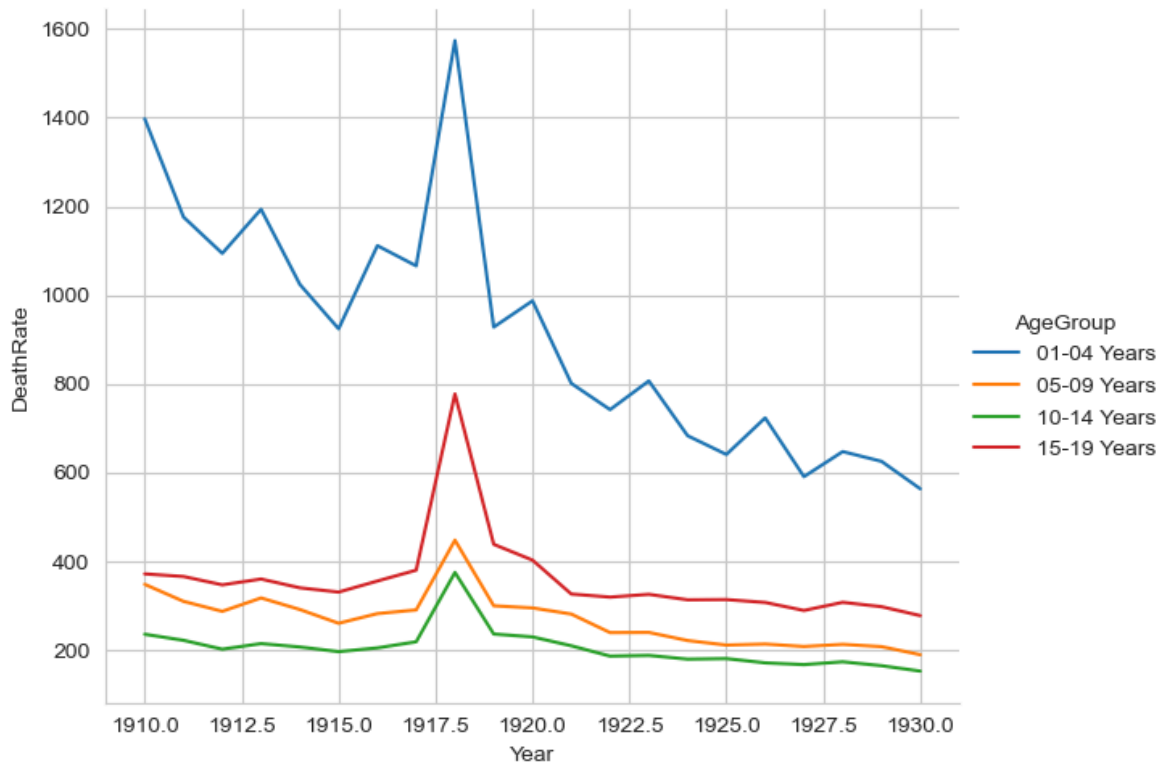
with pd.option_context('mode.use_inf_as_na', True):



```
In [13]: # Demonstrating the use of lineplot with Seaborn to plot mortality data, customi
sns.set_style('whitegrid')
sns.relplot(data=mortality_data.query('Year >= 1910 and Year <= 1930'),
            kind='line', x='Year', y='DeathRate', hue='AgeGroup', aspect=1.25)
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.
with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.
with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The figure layout has changed to tight
self.figure.tight_layout(*args, **kwargs)

```
Out[13]: <seaborn.axisgrid.FacetGrid at 0x2cdcb0a08d0>
```



```
In [14]: #Create a relational plot with Seaborn, grouping data by 'AgeGroup' and customiz
sns.set_style('darkgrid')
sns.relplot(data=mortality_data.query('Year >= 1910 and Year <= 1930'),
            kind='line', x='Year', y='DeathRate', hue='AgeGroup', aspect=1.25)
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

with pd.option_context('mode.use_inf_as_na', True):

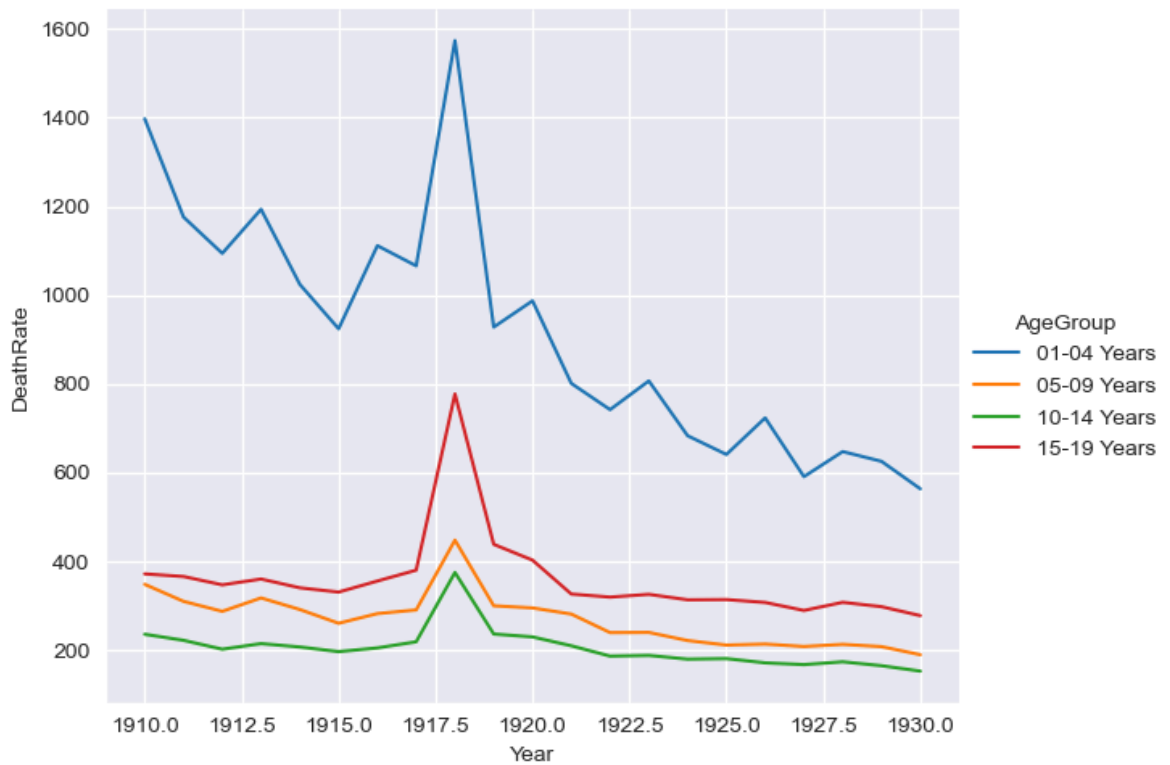
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

with pd.option_context('mode.use_inf_as_na', True):

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The figure layout has changed to tight

self._figure.tight_layout(*args, **kwargs)

```
Out[14]: <seaborn.axisgrid.FacetGrid at 0x2cdcb177090>
```



```
In [54]: # Opt for a minimalistic look with 'ticks' style, showing ticks on the x and y a
sns.set_style('ticks')
```

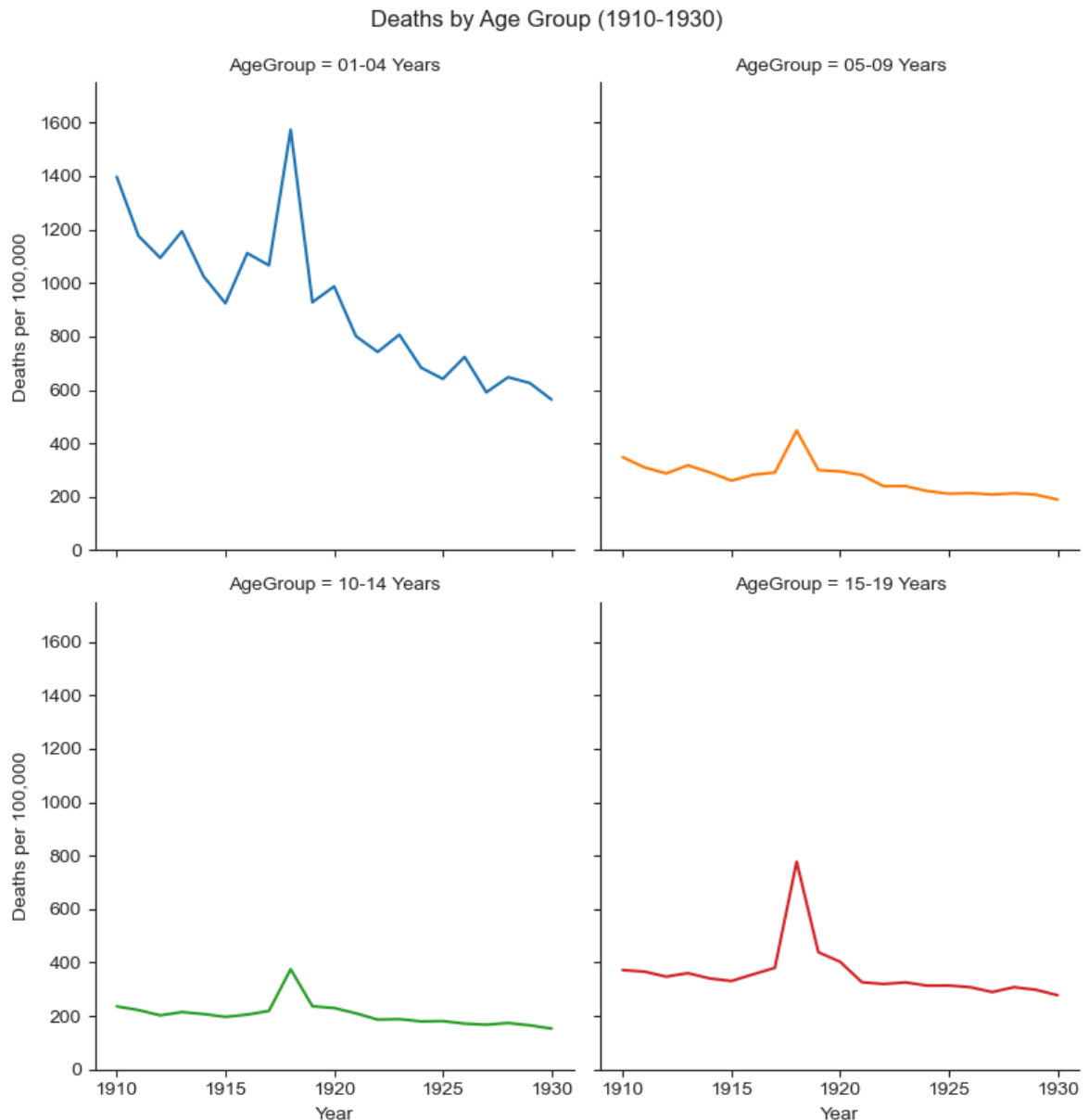
```
In [55]: # Created a relational plot for a specific time range (1910 to 1930), focusing o
g = sns.relplot(data=mortality_data.query('Year >= 1910 and Year <= 1930'),
kind='line', x='Year', y='DeathRate', hue='AgeGroup', legend=False,
col='AgeGroup', col_wrap=2, height=4)
g.fig.suptitle('Deaths by Age Group (1910-1930)', y=1.025)
g.set(ylabel='Deaths per 100,000',
xticks=[x for x in range(1910,1931,5)],
ylim=(0,1750))
```

```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
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rt inf values to NaN before operating instead.
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C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
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C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
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use_inf_as_na option is deprecated and will be removed in a future version. Conve
rt inf values to NaN before operating instead.
    with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: Th
e figure layout has changed to tight
    self._figure.tight_layout(*args, **kwargs)

```

Out[55]: <seaborn.axisgrid.FacetGrid at 0x2cdd3f9a1d0>



Relational plots

```
In [17]: #Plot mortality data focusing on years 1910 to 1930, showcasing how to filter data
# We use the 'query' method to select a subset of the data and 'kind=line' to create
sns.relplot(data=mortality_data, kind='line', x='Year', y='DeathRate',
            hue='AgeGroup')
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

```
with pd.option_context('mode.use_inf_as_na', True):
```

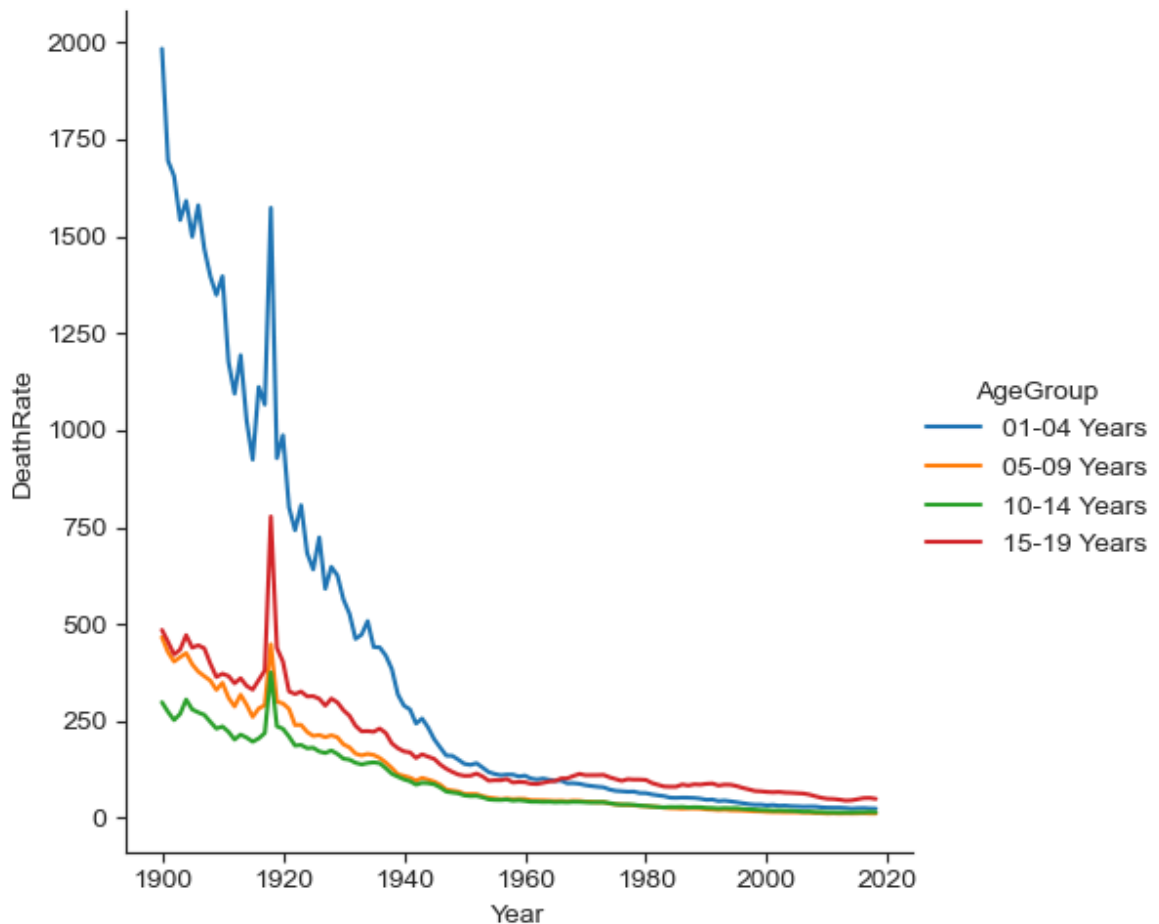
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

```
with pd.option_context('mode.use_inf_as_na', True):
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The figure layout has changed to tight

```
self.figure.tight_layout(*args, **kwargs)
```

```
Out[17]: <seaborn.axisgrid.FacetGrid at 0x2cdcc03fc90>
```



```
In [18]: # Use Seaborn to create a line plot showing mortality rates over time, customizing
sns.relplot(data=mortality_data, kind='line', x='Year', y='DeathRate', ci='sd')
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:848: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar='sd'` for the same effect.

```
func(*plot_args, **plot_kwargs)
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

```
with pd.option_context('mode.use_inf_as_na', True):
```

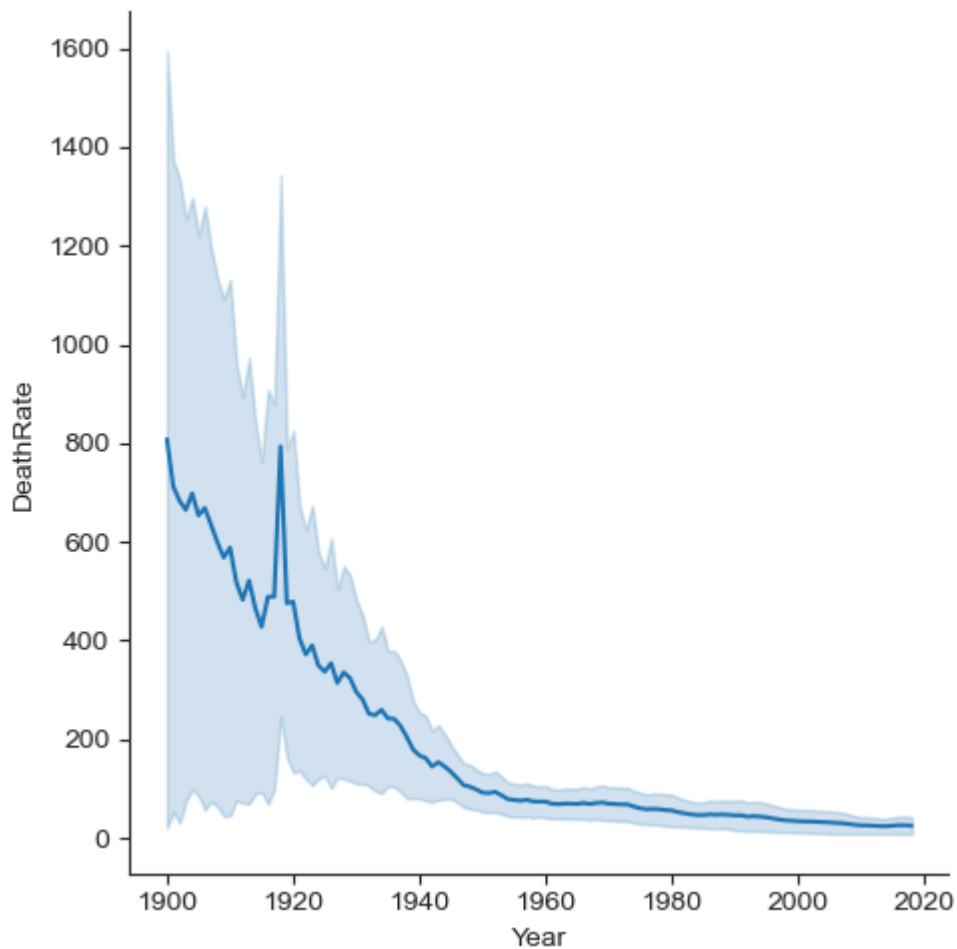
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

```
with pd.option_context('mode.use_inf_as_na', True):
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The figure layout has changed to tight

```
self._figure.tight_layout(*args, **kwargs)
```

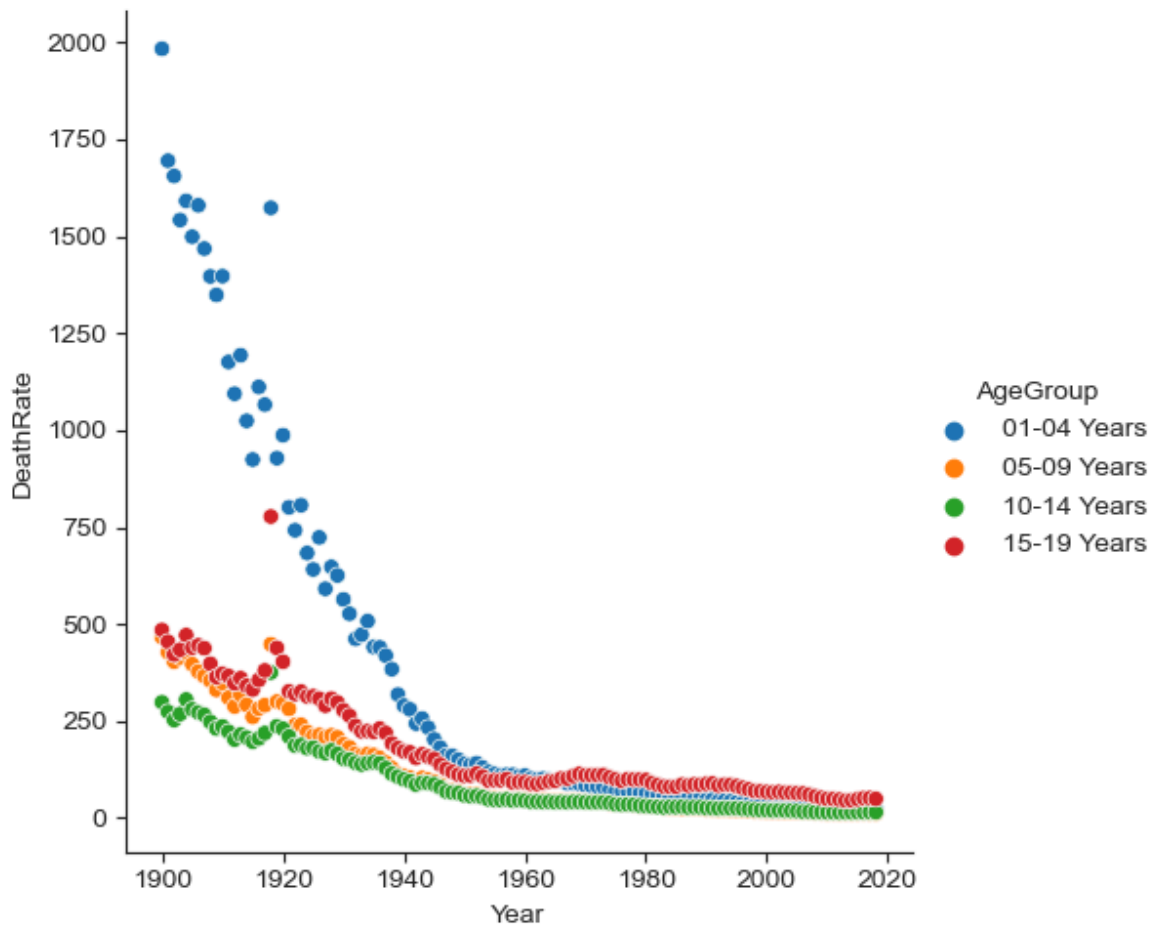
Out[18]: <seaborn.axisgrid.FacetGrid at 0x2cdcc01be90>



```
In [19]: # Display a scatter plot to examine the relationship between year and death rate
sns.relplot(data=mortality_data, kind='scatter', x='Year', y='DeathRate',
            hue='AgeGroup')
```

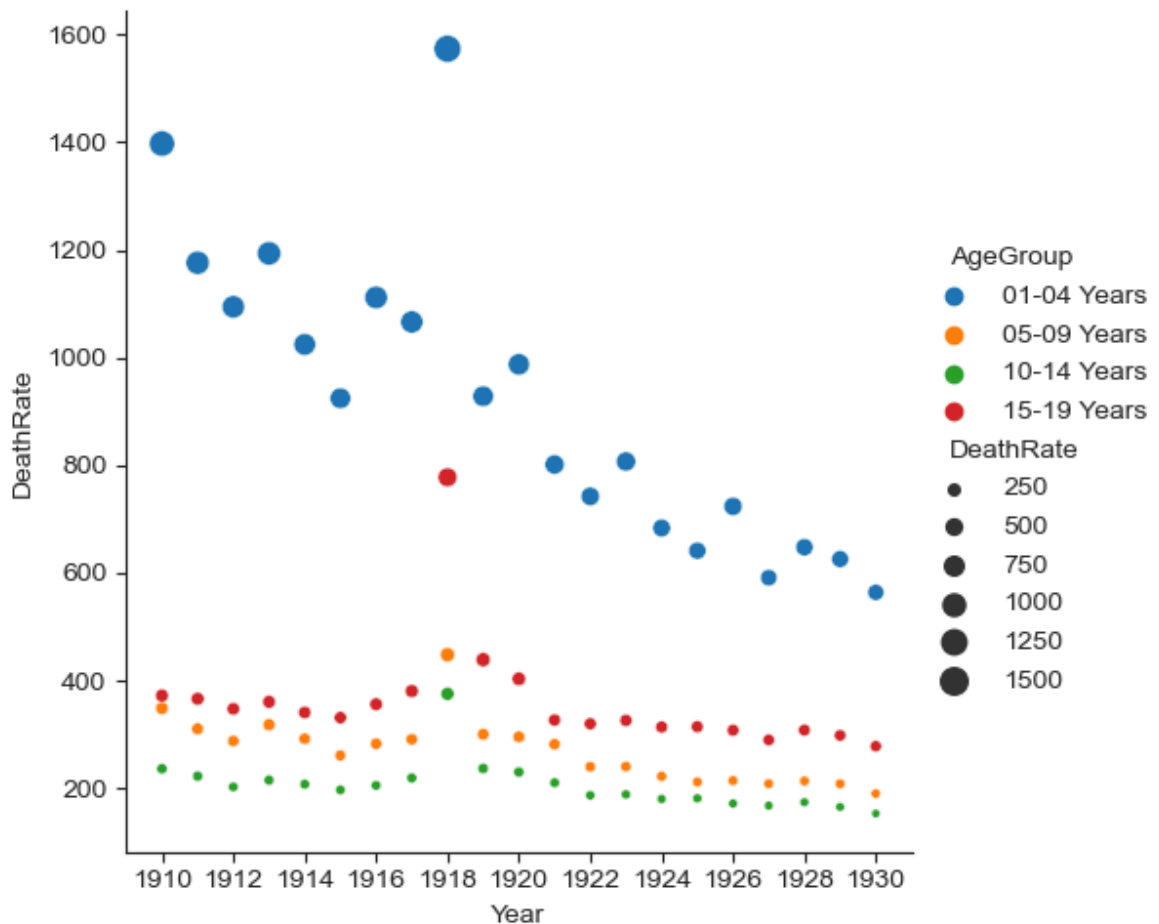
```
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The figure layout has changed to tight
self._figure.tight_layout(*args, **kwargs)
```

```
Out[19]: <seaborn.axisgrid.FacetGrid at 0x2cdcb9de0d0>
```



```
In [20]: # Focus on a specific time range (1910-1930) with a scatter plot, comparing death
g = sns.relplot(data=mortality_data.query('Year >= 1910 and Year <= 1930'),
kind='scatter', x='Year', y='DeathRate', hue='AgeGroup',
size='DeathRate', sizes=(10,100))
for ax in g.axes.flat:
    ax.set(xticks=[x for x in range(1910,1931,2)])
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The figure layout has changed to tight
self._figure.tight_layout(*args, **kwargs)



Categorical plots

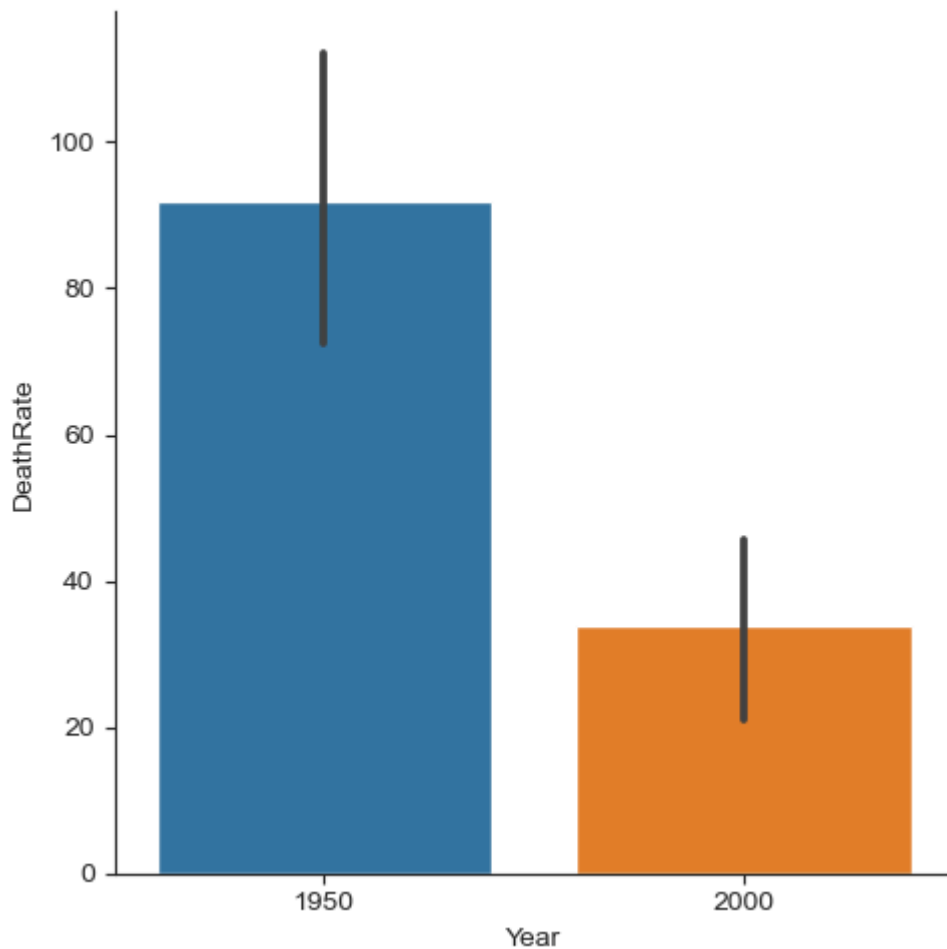
In [21]: `#Changed cell 20 so it uses a specific method to create the bar plot, and chang
sns.catplot(data=mortality_data.query('Year in (1950,2000)'),
kind='bar', x='Year', y='DeathRate', ci=75)`

C:\Users\Dell\AppData\Local\Temp\ipykernel_10564\3672341933.py:1: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar=('ci', 75)` for the same effect.

```
sns.catplot(data=mortality_data.query('Year in (1950,2000)'),
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: Th
e figure layout has changed to tight
self._figure.tight_layout(*args, **kwargs)
```

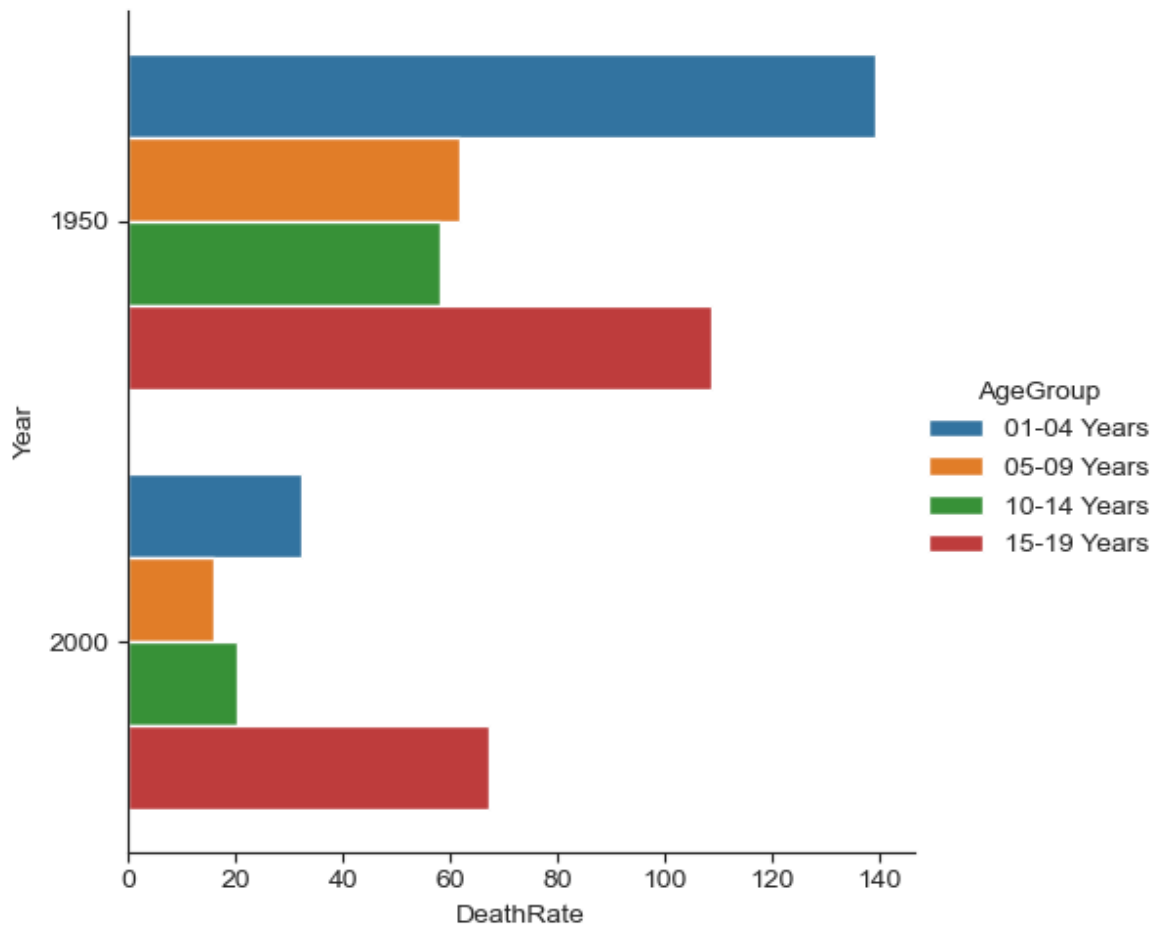
Out[21]: `<seaborn.axisgrid.FacetGrid at 0x2cdcc0c0850>`



```
In [22]: # Create a horizontal bar plot using Seaborn's catplot to visualize the distribu
# The 'query' function filters the dataset for these years, and 'orient='h'' mak
# 'x' is set to 'DeathRate' to define the length of the bars, 'y' to 'Year' to c
sns.catplot(data=mortality_data.query('Year in (1950,2000)'),
            kind='bar', x='DeathRate', y='Year', hue='AgeGroup', orient='h')
```

```
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: Th
e figure layout has changed to tight
self._figure.tight_layout(*args, **kwargs)
```

```
Out[22]: <seaborn.axisgrid.FacetGrid at 0x2cdcd3d0d10>
```

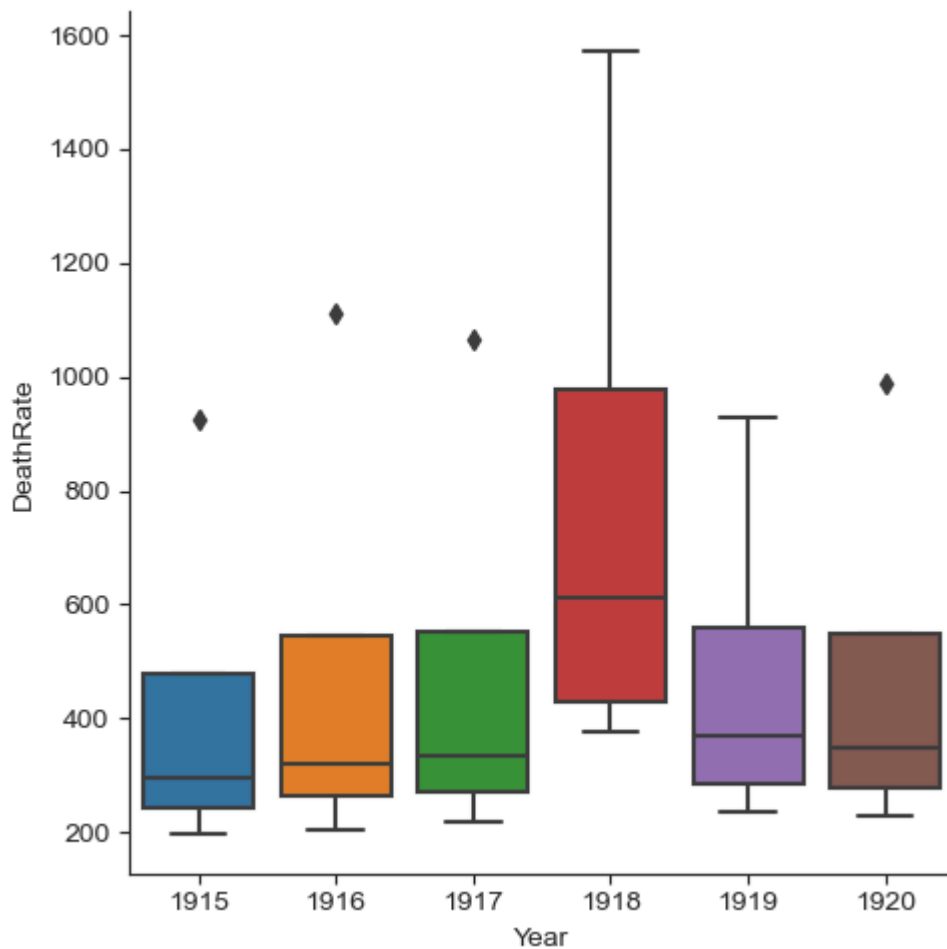


```
In [23]: # Generate a box plot using Seaborn's catplot to examine the distribution of death rates
# The 'query' method is used to filter the dataset to include only the years between 1915 and 1920
# The plot categorizes data by 'Year' on the x-axis and shows the distribution of death rates
# Box plots are useful for visualizing the central tendency, dispersion, and outliers
sns.catplot(data=mortality_data.query('Year >= 1915 and Year <= 1920'),
            kind='box', x='Year', y='DeathRate')
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The figure layout has changed to tight

```
self._figure.tight_layout(*args, **kwargs)
```

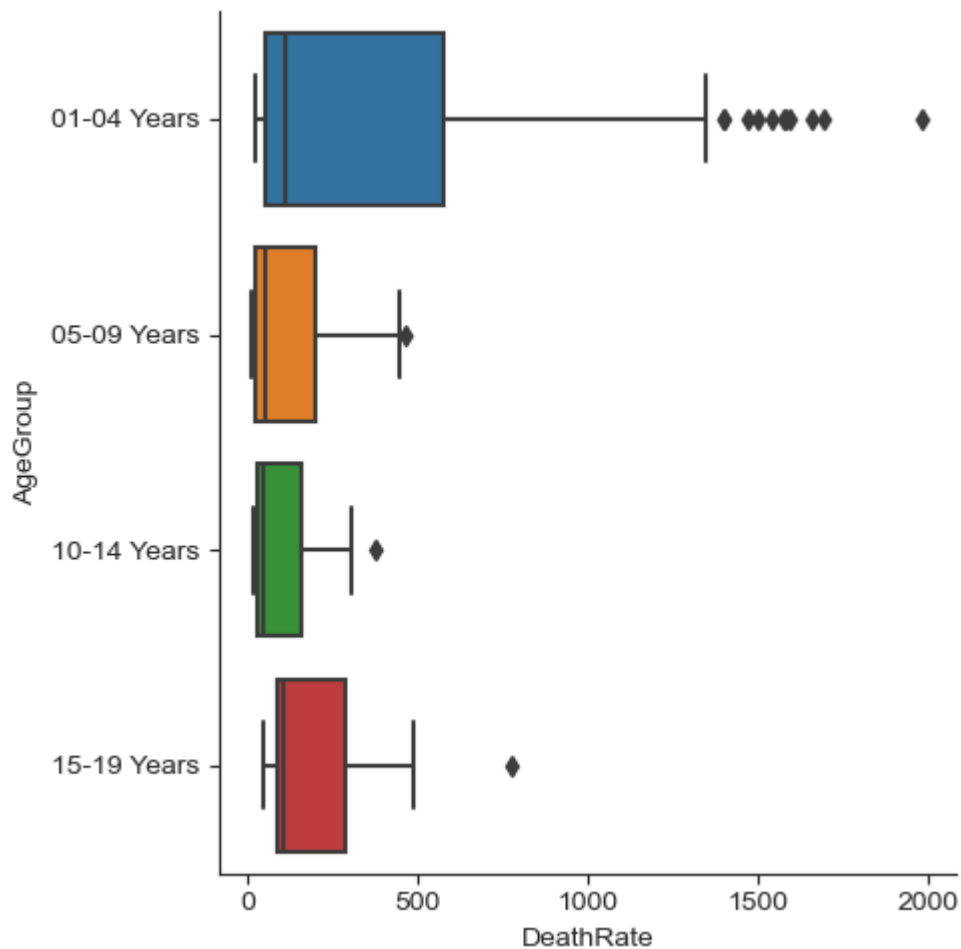
```
Out[23]: <seaborn.axisgrid.FacetGrid at 0x2cdcab8c310>
```



```
In [24]: # Generate a box plot using Seaborn's catplot to examine the distribution of dea
# The 'query' method is used to filter the dataset to include only the years bet
# The plot categorizes data by 'Year' on the x-axis and shows the distribution o
# Box plots are useful for visualizing the central tendency, dispersion, and out
sns.catplot(data=mortality_data,
            kind='box', x='DeathRate', y='AgeGroup', orient='h')
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The figure layout has changed to tight
self._figure.tight_layout(*args, **kwargs)

```
Out[24]: <seaborn.axisgrid.FacetGrid at 0x2cdcabb3d0>
```



Distribution plots

```
In [25]: # Generate a histogram using Seaborn's displot to visualize the overall distribu
# The 'kind' parameter is set to 'hist' to specify that a histogram should be cr
# This plot provides insight into the frequency distribution of death rates, all
sns.displot(data=mortality_data, kind='hist', x='DeathRate')
```

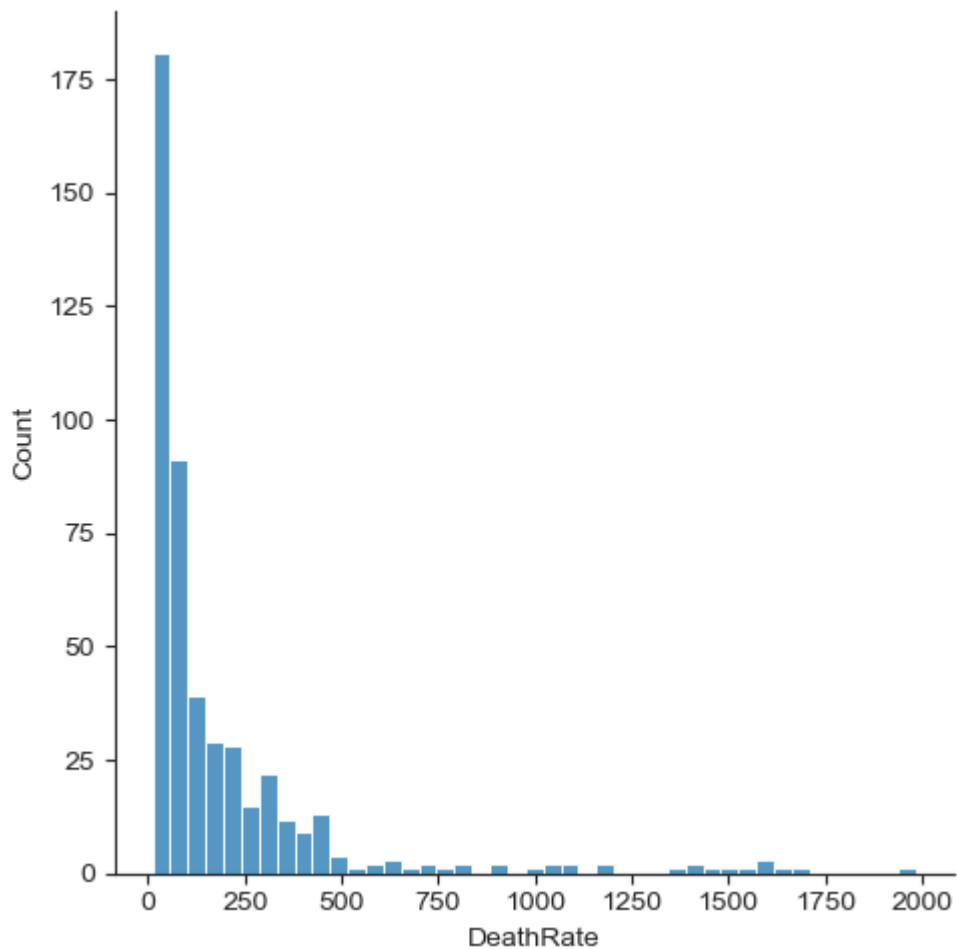
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

with pd.option_context('mode.use_inf_as_na', True):

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The figure layout has changed to tight

self.figure.tight_layout(*args, **kwargs)

```
Out[25]: <seaborn.axisgrid.FacetGrid at 0x2cdc9234510>
```



```
In [26]: #Change cell 25 so the histogram uses twice as many bins as it currently does for
sns.displot(data=mortality_data, kind='hist', x='DeathRate', bins=16)
```

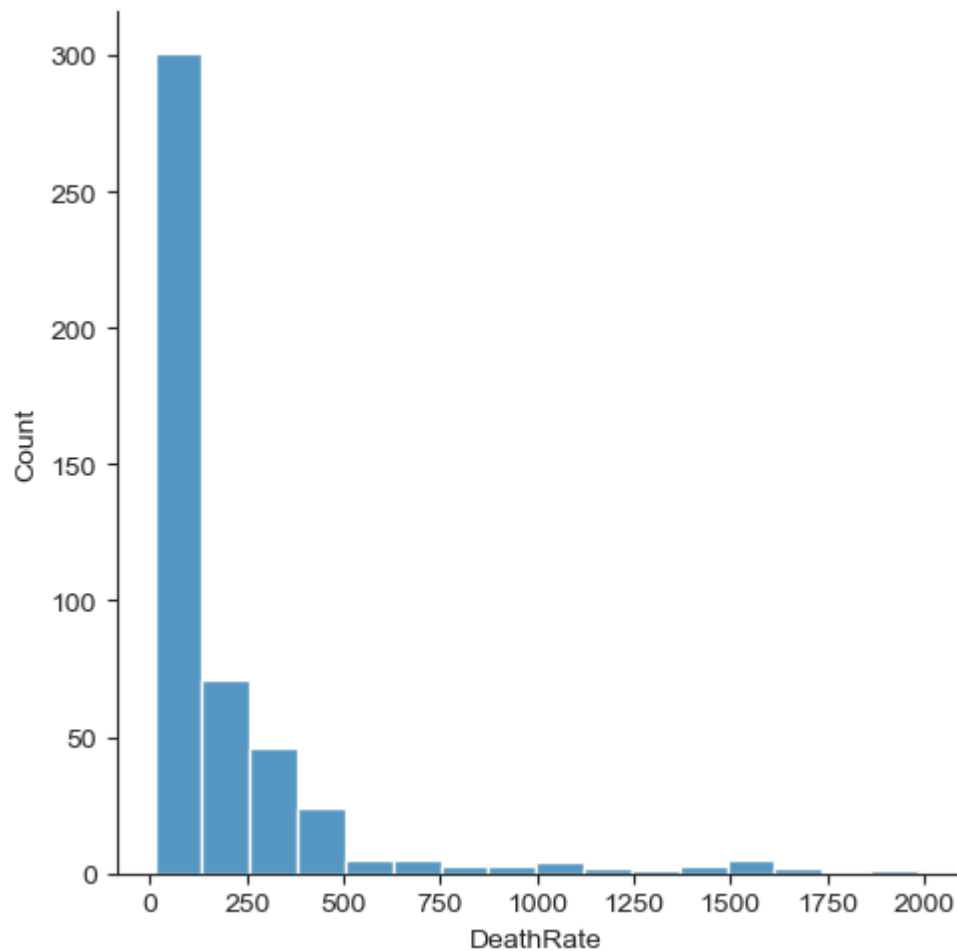
```
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
rt inf values to NaN before operating instead.
```

```
    with pd.option_context('mode.use_inf_as_na', True):
```

```
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: Th
e figure layout has changed to tight
```

```
    self._figure.tight_layout(*args, **kwargs)
```

```
Out[26]: <seaborn.axisgrid.FacetGrid at 0x2cdcd860890>
```



```
In [27]: # Create a Kernel Density Estimate (KDE) plot with Seaborn to visualize the prob  
sns.displot(data=mortality_data, kind='kde', x='DeathRate')
```

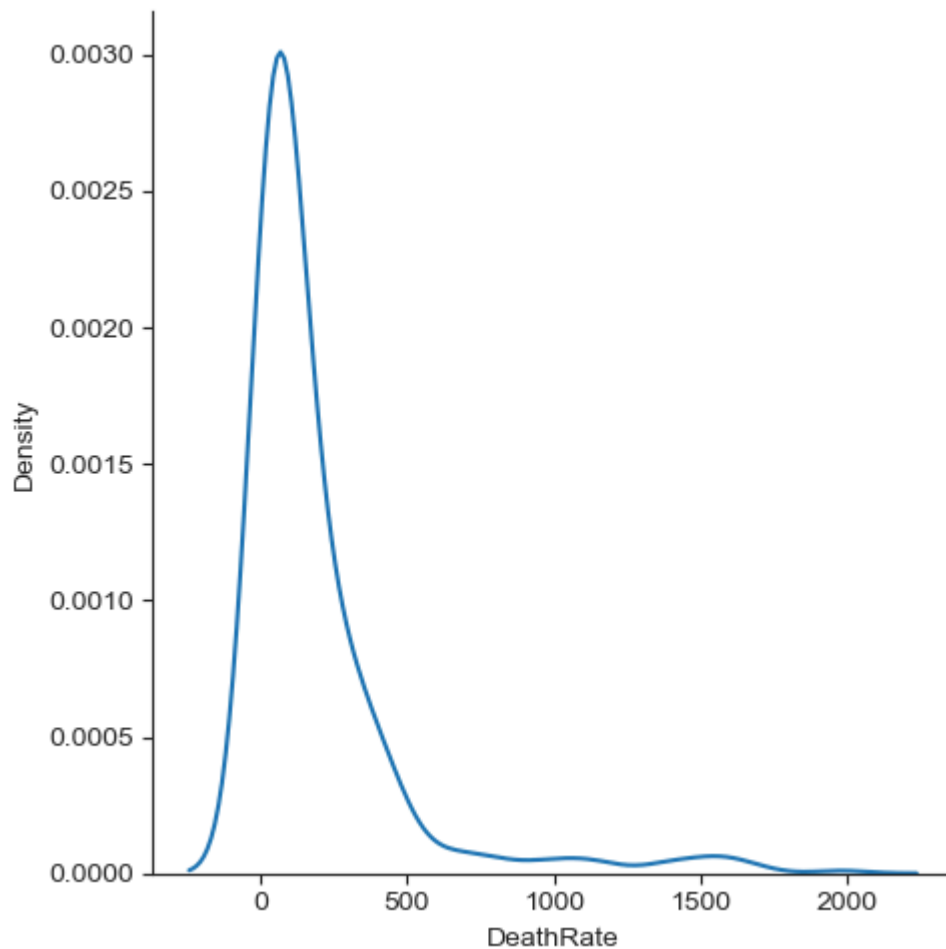
```
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:  
use_inf_as_na option is deprecated and will be removed in a future version. Conve  
rt inf values to NaN before operating instead.
```

```
    with pd.option_context('mode.use_inf_as_na', True):
```

```
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: Th  
e figure layout has changed to tight
```

```
    self._figure.tight_layout(*args, **kwargs)
```

```
Out[27]: <seaborn.axisgrid.FacetGrid at 0x2cdcd3d68d0>
```



```
In [28]: # Generate an Empirical Cumulative Distribution Function (ECDF) plot to visualize
sns.displot(data=mortality_data, kind='ecdf', x='DeathRate', hue='AgeGroup')
```

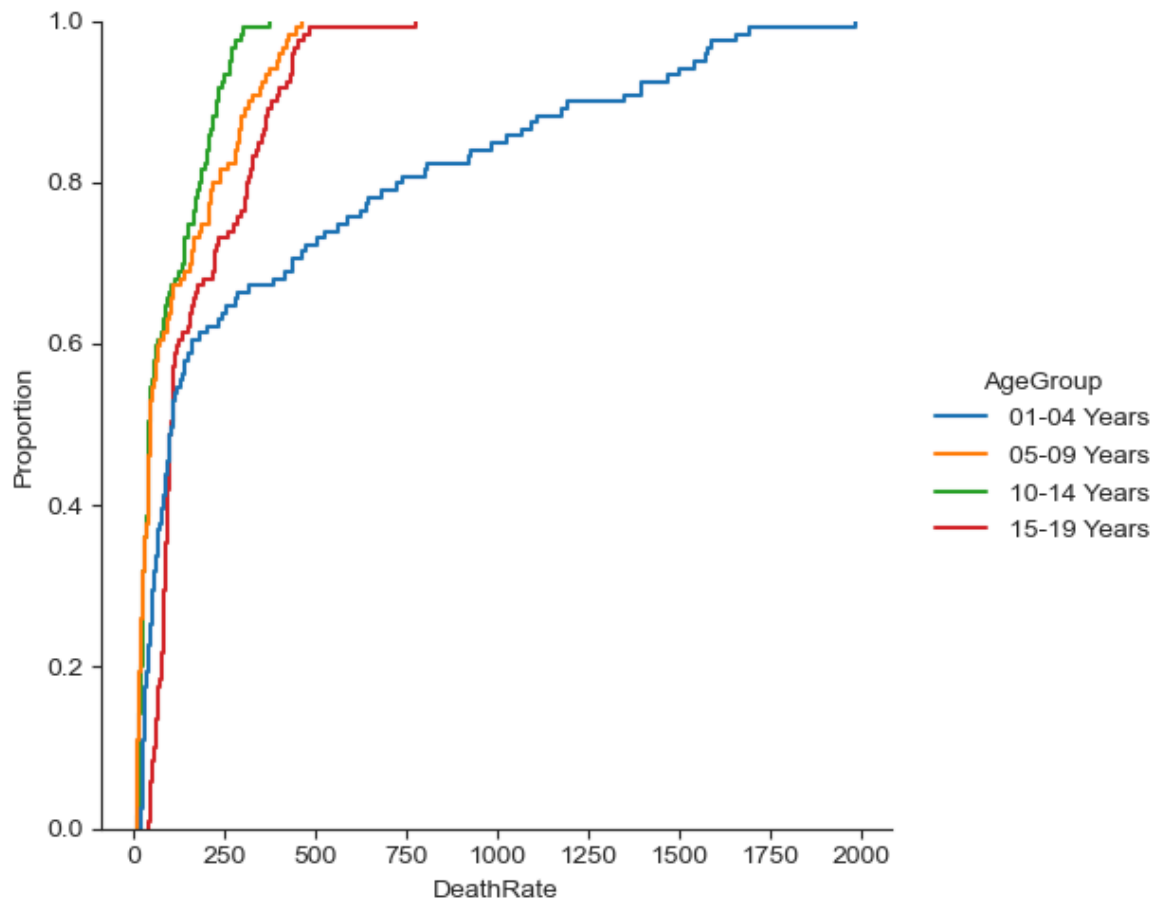
```
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Convert
inf values to NaN before operating instead.
```

```
    with pd.option_context('mode.use_inf_as_na', True):
```

```
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The
figure layout has changed to tight
```

```
    self._figure.tight_layout(*args, **kwargs)
```

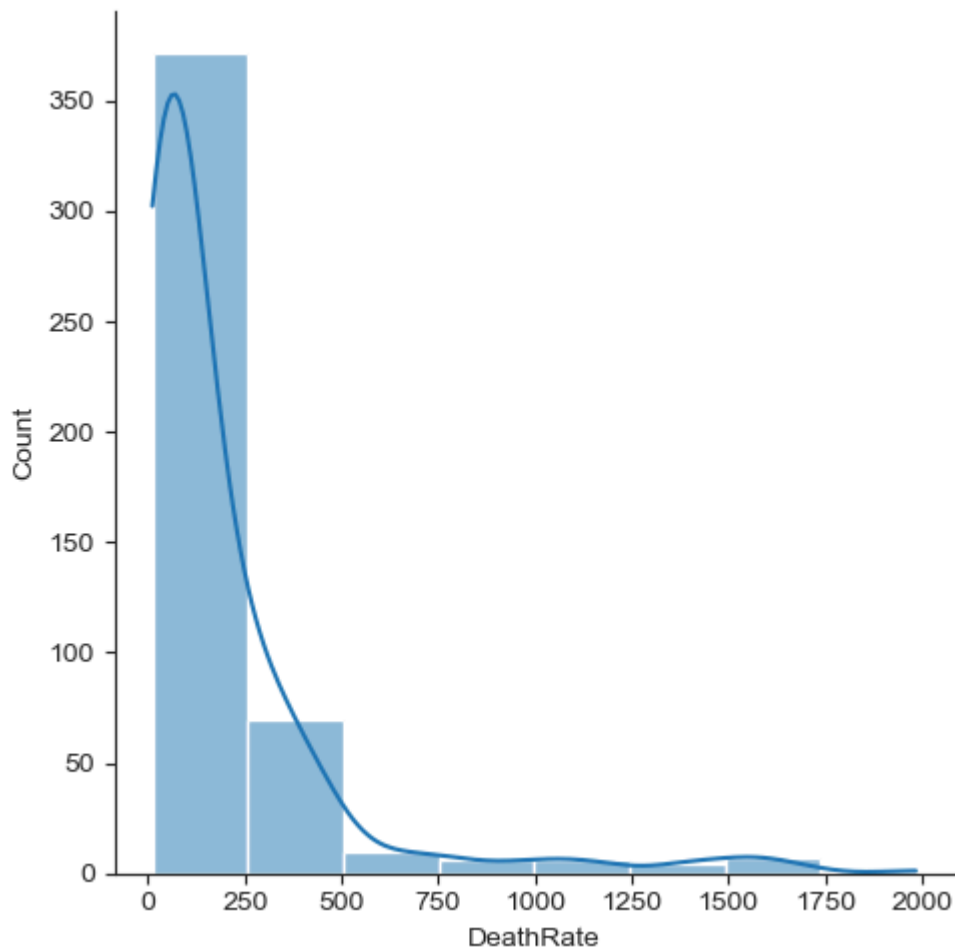
```
Out[28]: <seaborn.axisgrid.FacetGrid at 0x2cdcd90d9d0>
```

```
In [29]: # Create a histogram with an overlaid Kernel Density Estimate (KDE) to analyze
sns.displot(data=mortality_data, kind='hist',
            x='DeathRate', kde=True, bins=8)
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.
 with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The figure layout has changed to tight
 self._figure.tight_layout(*args, **kwargs)

```
Out[29]: <seaborn.axisgrid.FacetGrid at 0x2cdcd8e3bd0>
```



```
In [30]: # Generate filled KDE plots for death rates by age group, organized into a 2-col
sns.displot(data=mortality_data, kind='kde', x='DeathRate', hue='AgeGroup',
            fill=True, col='AgeGroup', col_wrap=2, height=3, legend=False)
```

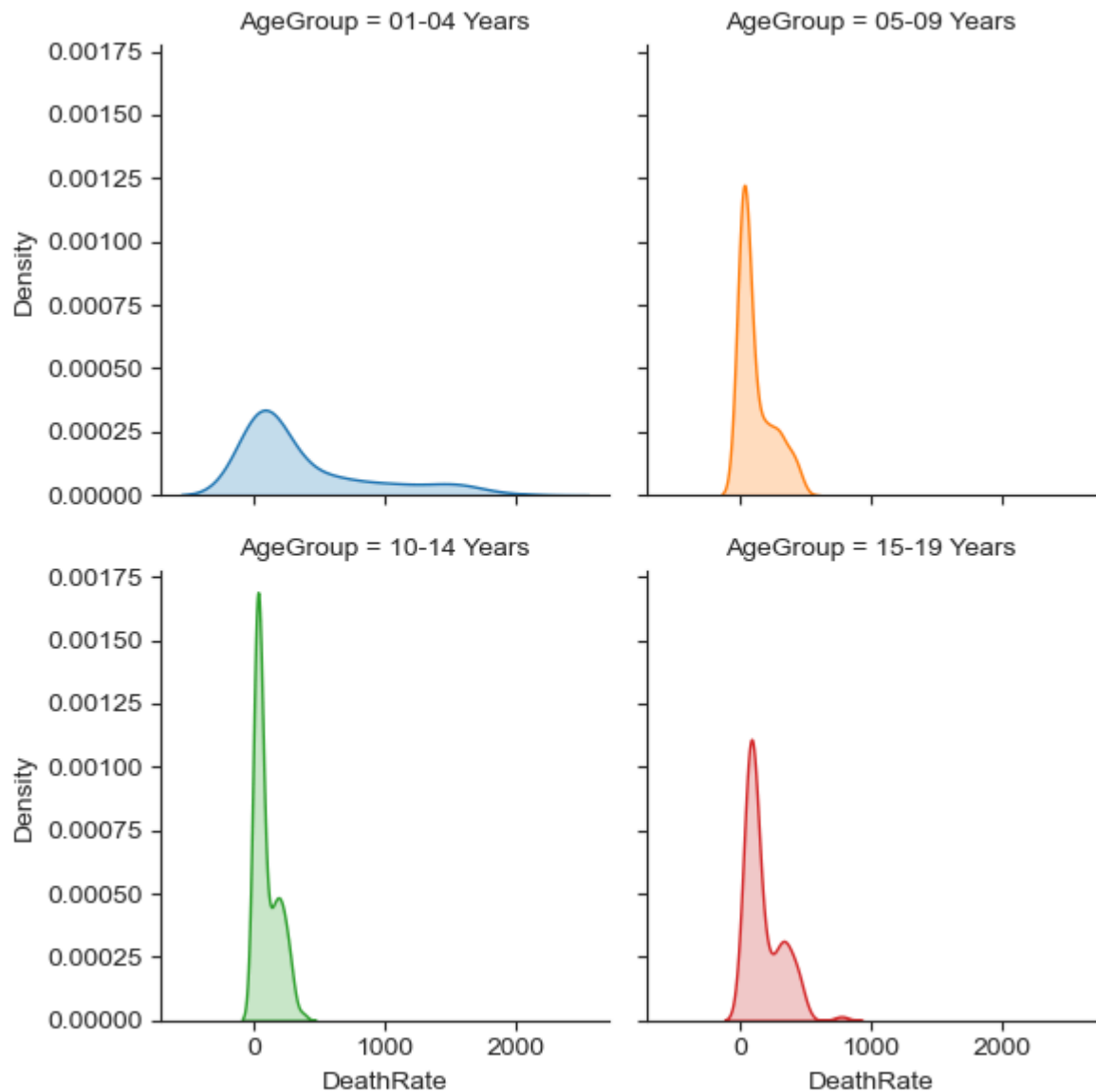
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

with pd.option_context('mode.use_inf_as_na', True):

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The figure layout has changed to tight

self._figure.tight_layout(*args, **kwargs)

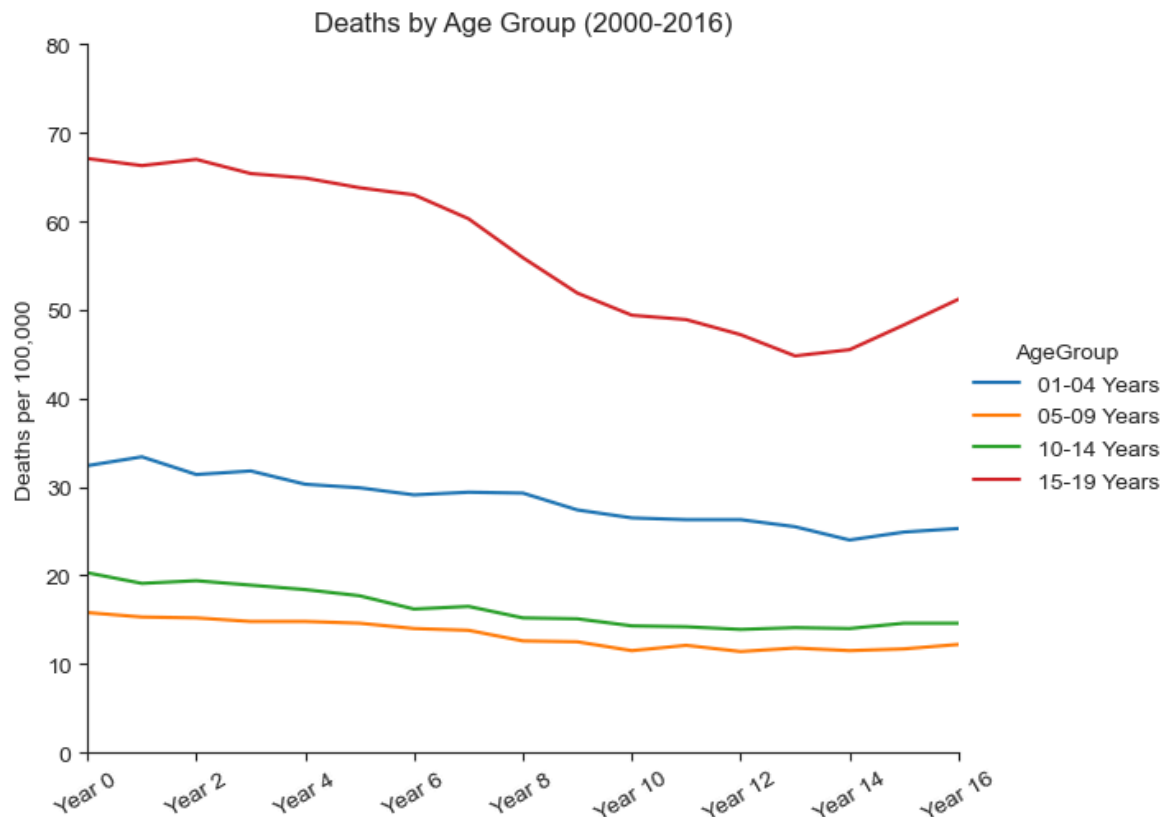
```
Out[30]: <seaborn.axisgrid.FacetGrid at 0x2cdcfb28e50>
```



Other techniques for enhancing a plot

```
In [31]: # Create a line plot to show death rates by age group from 2000 to 2016, customi
g = sns.relplot(
    data=mortality_data,
    kind='line', x='Year', y='DeathRate', hue='AgeGroup', aspect=1.25)
for ax in g.axes.flat:
    ax.set_title('Deaths by Age Group (2000-2016)')
    ax.set_xlabel('')
    ax.set_xticks([x for x in range(2000,2017,2)])
    ax.set_xticklabels(['Year ' + str(x) for x in range(0,17,2)])
    ax.tick_params('x', labelrotation=30)
    ax.set_ylabel('Deaths per 100,000')
    ax.set_xlim(2000,2016)
    ax.set_ylim(0,80)
```

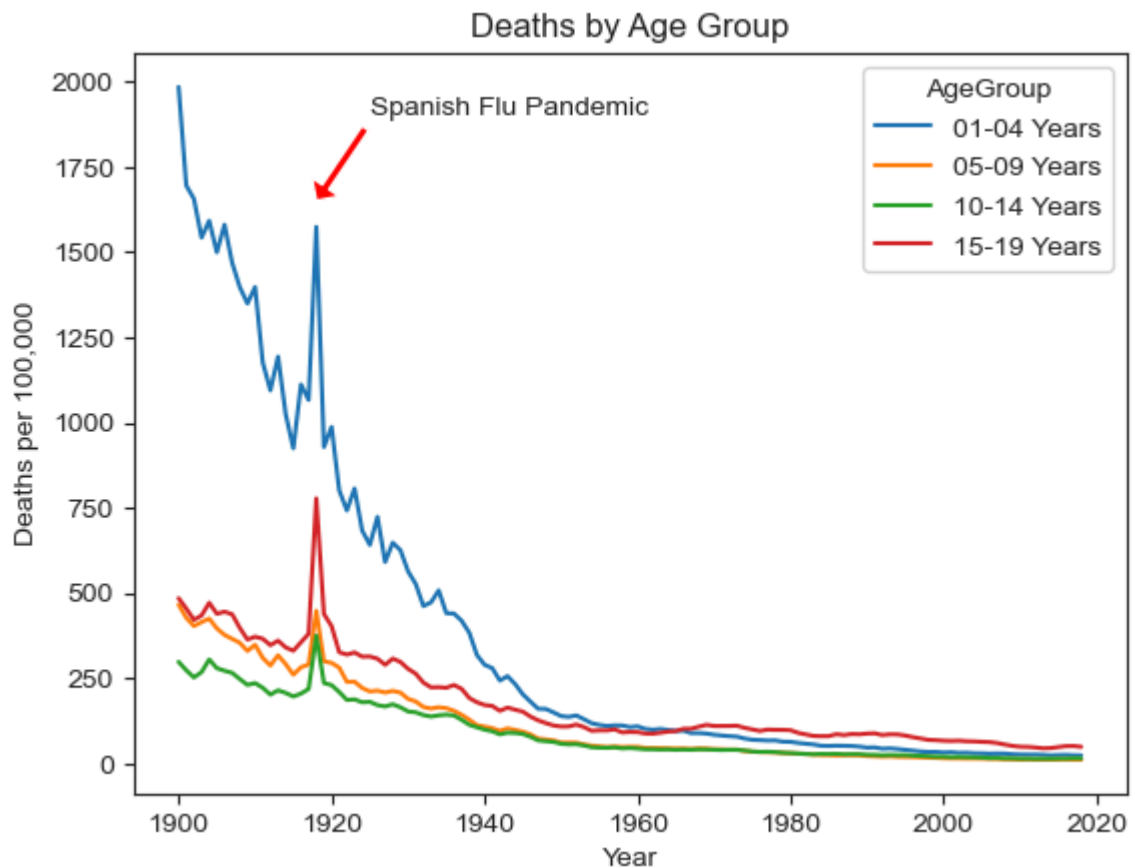
```
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
rt inf values to NaN before operating instead.
    with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
rt inf values to NaN before operating instead.
    with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: Th
e figure layout has changed to tight
    self.figure.tight_layout(*args, **kwargs)
```



```
In [32]: # Plot death rates by age group over years with annotations highlighting the Spa
ax = sns.lineplot(data=mortality_data,
                  x='Year', y='DeathRate', hue='AgeGroup')
ax.set_title('Deaths by Age Group')
ax.set_ylabel('Deaths per 100,000')
ax.annotate(text='Spanish Flu Pandemic',
           xy=(1918, 1650), xytext=(1925, 1900),
           arrowprops=dict(facecolor='red', width=3, headwidth=12, headlength=6))
```

```
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
rt inf values to NaN before operating instead.
    with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
rt inf values to NaN before operating instead.
    with pd.option_context('mode.use_inf_as_na', True):
```

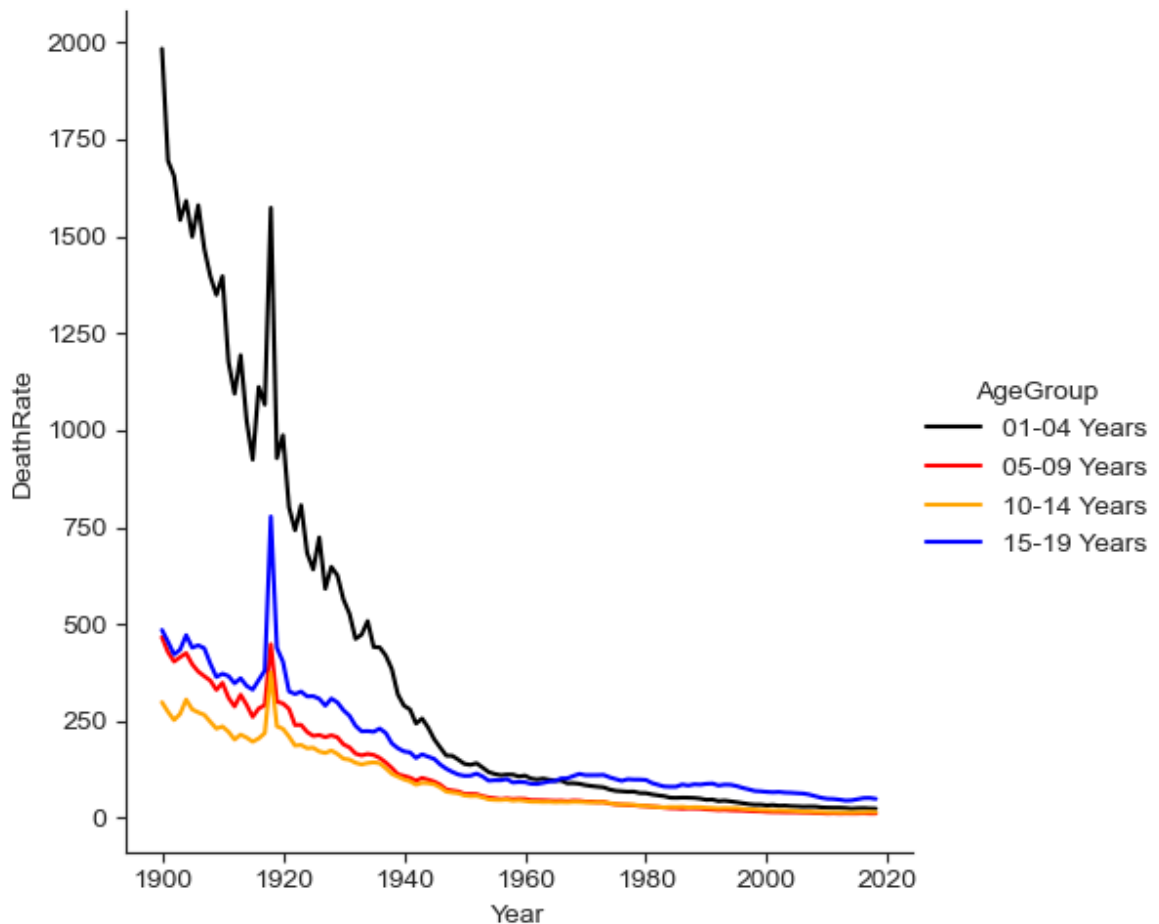
```
Out[32]: Text(1925, 1900, 'Spanish Flu Pandemic')
```



```
In [33]: # Set a custom color palette and create a line plot to visualize death rates by
          colors = ['black', 'red', 'orange', 'blue']
          sns.set_palette(sns.color_palette(colors))
          sns.relplot(data=mortality_data, kind='line',
                      x='Year', y='DeathRate', hue='AgeGroup')
```

```
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
rt inf values to NaN before operating instead.
  with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
rt inf values to NaN before operating instead.
  with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: Th
e figure layout has changed to tight
  self._figure.tight_layout(*args, **kwargs)
```

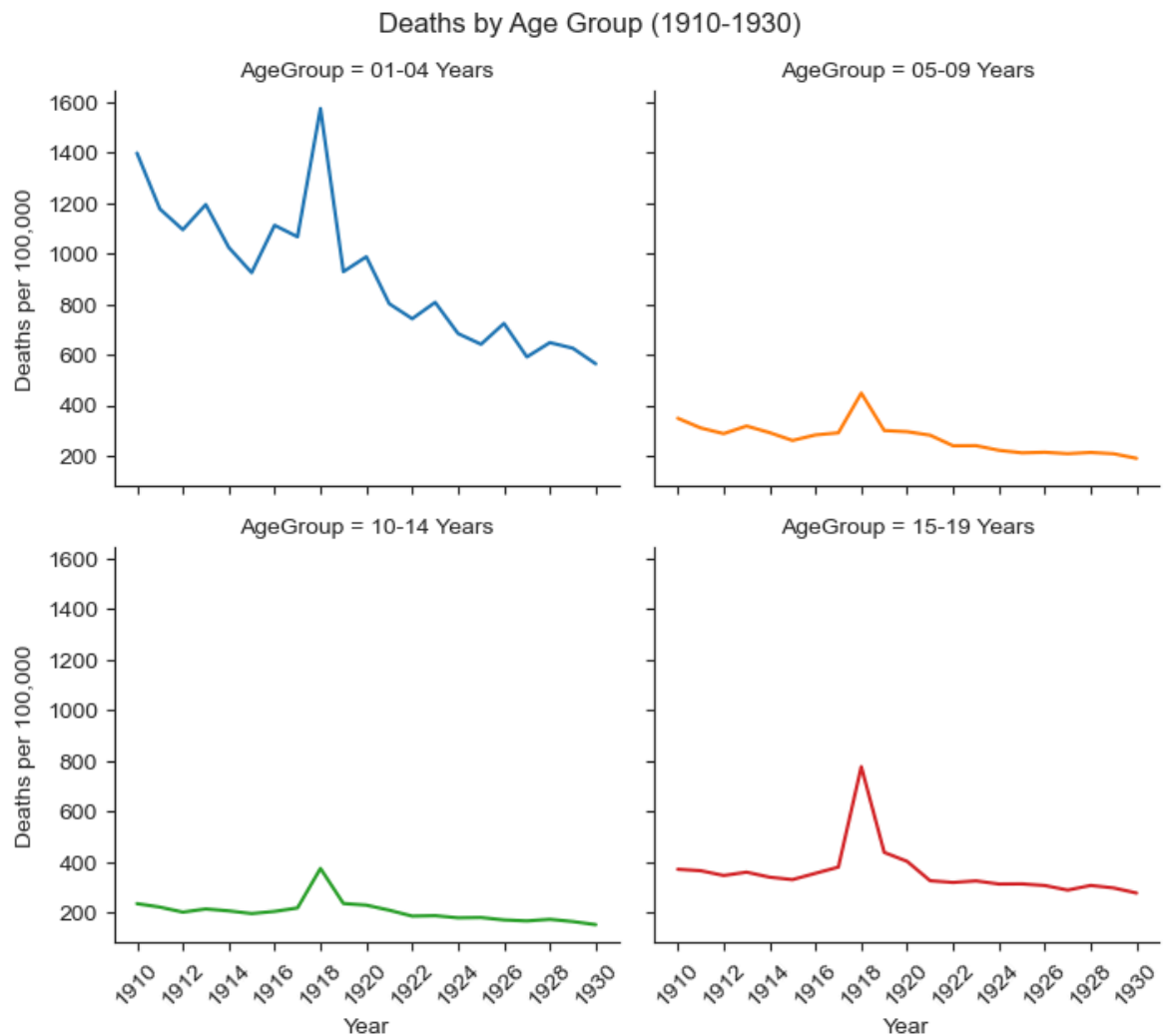
```
Out[33]: <seaborn.axisgrid.FacetGrid at 0x2cdd04ee510>
```



In [35]: `# Reset the color palette to 'tab10' for subsequent plots, offering a diverse an
sns.set_palette(sns.color_palette('tab10'))`

In [36]: `# Create segmented line plots for death rates by age group between 1910 and 1930
g = sns.relplot(
 data=mortality_data.query('Year >= 1910 and Year <= 1930'),
 kind='line', x='Year', y='DeathRate', hue='AgeGroup', legend=False,
 col='AgeGroup', col_wrap=2, height=3, aspect=1.2)
g.fig.suptitle('Deaths by Age Group (1910-1930)', y=1.025)
for ax in g.axes.flat:
 ax.set_ylabel('Deaths per 100,000')
 ax.set_xticks([x for x in range(1910, 1931, 2)])
 ax.tick_params('x', labelrotation=45)`

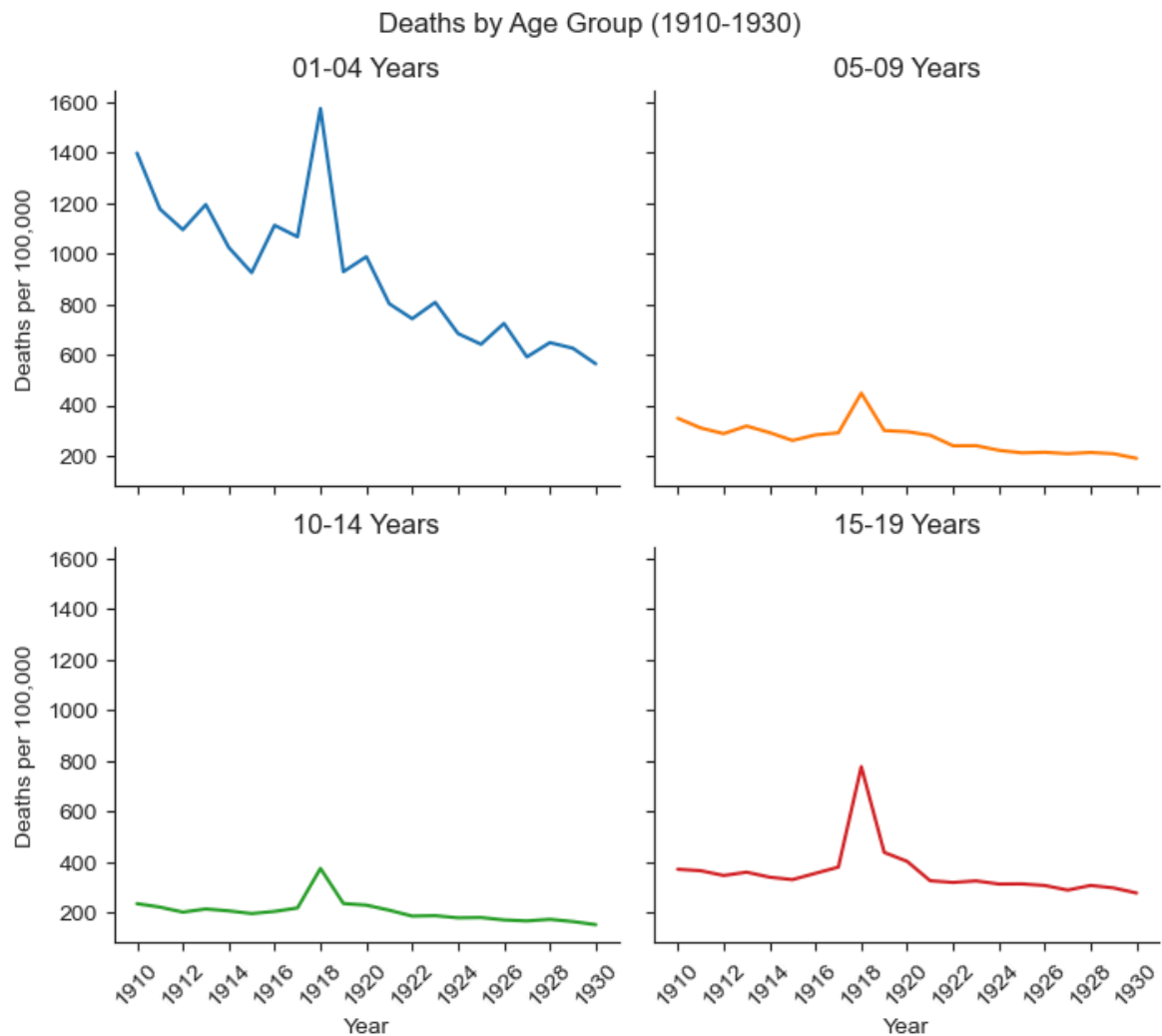
```
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
rt inf values to NaN before operating instead.
    with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
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    with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
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    with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
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    with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
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    with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
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C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
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C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
rt inf values to NaN before operating instead.
    with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: Th
e figure layout has changed to tight
    self._figure.tight_layout(*args, **kwargs)
```



```
In [37]: # Generate line plots for each age group's death rates from 1910 to 1930, custom
g = sns.relplot(
    data=mortality_data.query('Year >= 1910 and Year <= 1930'),
    kind='line', x='Year', y='DeathRate', hue='AgeGroup', legend=False,
    col='AgeGroup', col_wrap=2, height=3, aspect=1.2)
g.fig.suptitle('Deaths by Age Group (1910-1930)', y=1.025)
ageGroups = mortality_data['AgeGroup'].drop_duplicates().tolist()
for index, ax in enumerate(g.axes.flat):
    ax.set_title(ageGroups[index])
    ax.set_ylabel('Deaths per 100,000')
    ax.set_xticks([x for x in range(1910, 1931, 2)])
    ax.tick_params('x', labelrotation=45)
```



```
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
rt inf values to NaN before operating instead.
    with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
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C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
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    with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
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    with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
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C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
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    with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
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    with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
rt inf values to NaN before operating instead.
    with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: Th
e figure layout has changed to tight
    self._figure.tight_layout(*args, **kwargs)
```



```
In [38]: # import a Matplotlib module and set the figure size
import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(10,7.5))

# create the line plot
sns.lineplot(data=mortality_data, x='Year', y='DeathRate', hue='AgeGroup')

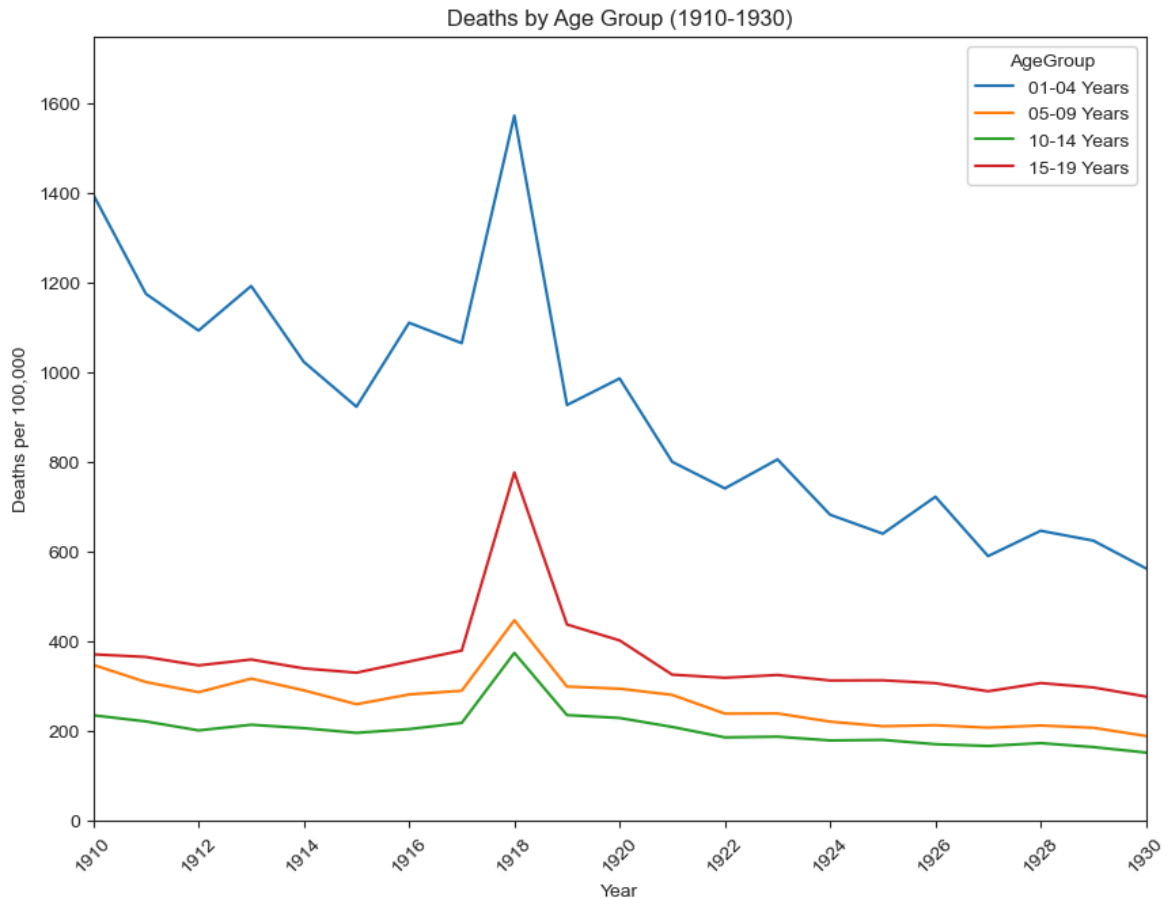
# use the ax and fig objects to enhance and save the plot
ax.set(title='Deaths by Age Group (1910-1930)',
       ylabel='Deaths per 100,000',
       xticks=[x for x in range(1910, 1931, 2)],
       xlim=(1910,1930), ylim=(0,1750)),
       ax.tick_params('x', labelrotation=45)
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

with pd.option_context('mode.use_inf_as_na', True):

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

with pd.option_context('mode.use_inf_as_na', True):

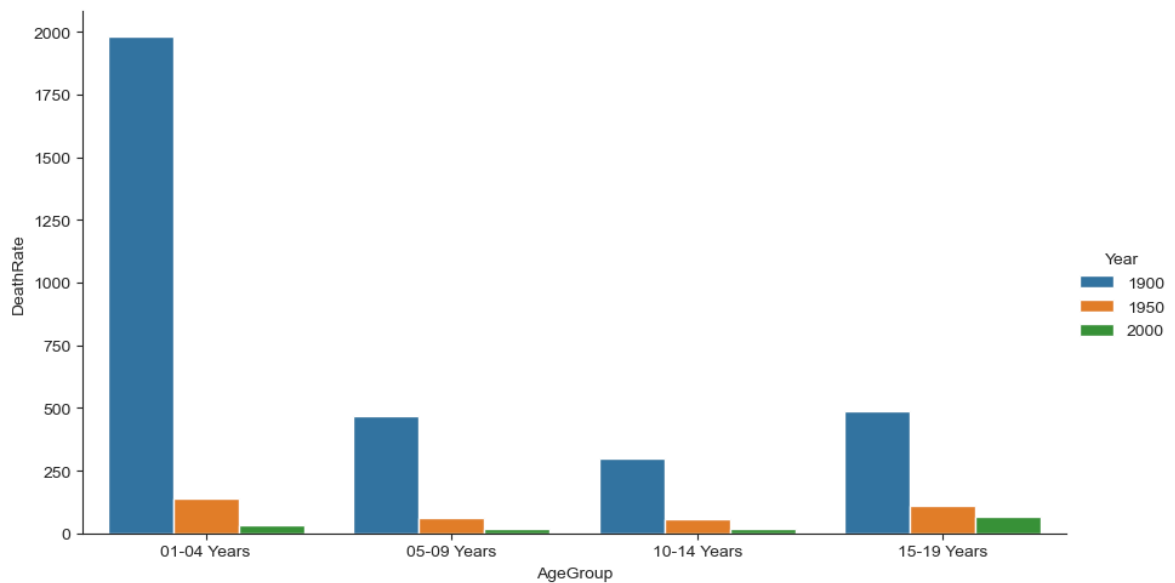


Create a vertical bar plot that shows the deaths rates for the four age groups for the years 1900, 1950, and 2000, and note that the bars represent the average death rates for the three years. Now, modify the plot so it shows the death rates for each of the three years, and increase the width of the plot so it's 1.8 times the height of the plot.

```
In [46]: # Create a bar plot comparing death rates across different age groups for the years 1900, 1950, and 2000
sns.catplot(data=mortality_data.query('Year in (1900,1950, 2000)'),
            kind='bar', x='AgeGroup', y='DeathRate', hue='Year', aspect=1.8)
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The figure layout has changed to tight
self._figure.tight_layout(*args, **kwargs)

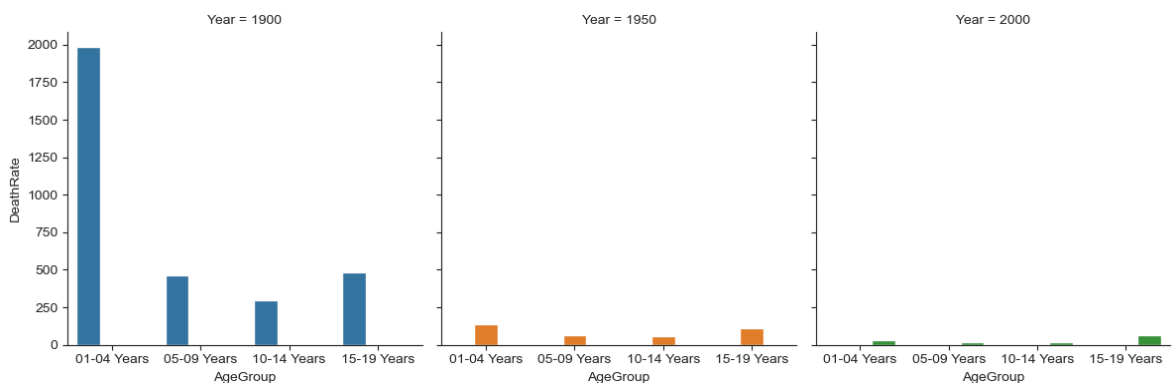
```
Out[46]: <seaborn.axisgrid.FacetGrid at 0x2cdd3d2b4d0>
```



```
In [48]: # Generate separate bar plots for death rates by age group in 1900, 1950, and 2000
sns.catplot(data=mortality_data.query('Year in (1900,1950, 2000)'),
            kind='bar', x='AgeGroup', y='DeathRate', hue='Year', height=4, col=
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The figure layout has changed to tight
self._figure.tight_layout(*args, **kwargs)

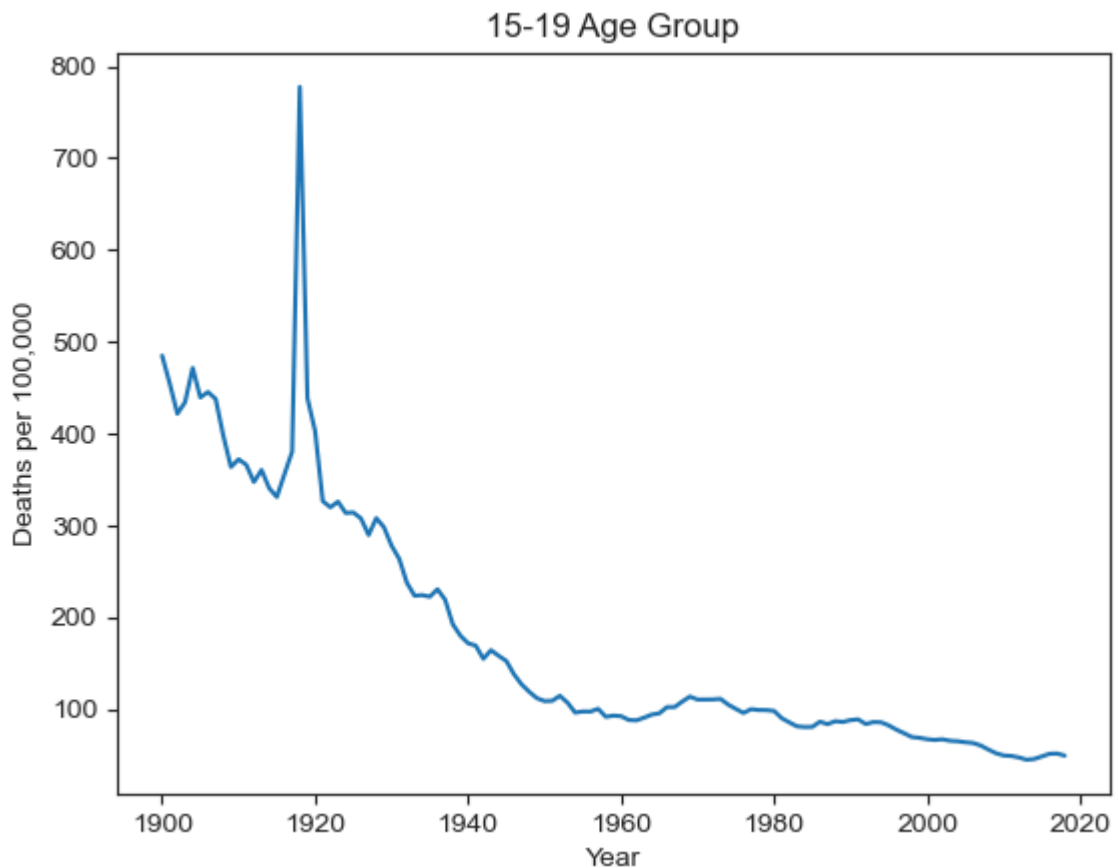
Out[48]: <seaborn.axisgrid.FacetGrid at 0x2cdd3fb9c90>



```
In [56]: # Plot death rates over years for the "15-19 Years" age group, setting the plot
ax = sns.lineplot(data=mortality_data.query('AgeGroup == "15-19 Years"'), x='Year', y='DeathRate')
ax.set(title = '15-19 Age Group', ylabel='Deaths per 100,000')
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.
with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn_oldcore.py:1119: FutureWarning: use_inf_as_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.
with pd.option_context('mode.use_inf_as_na', True):

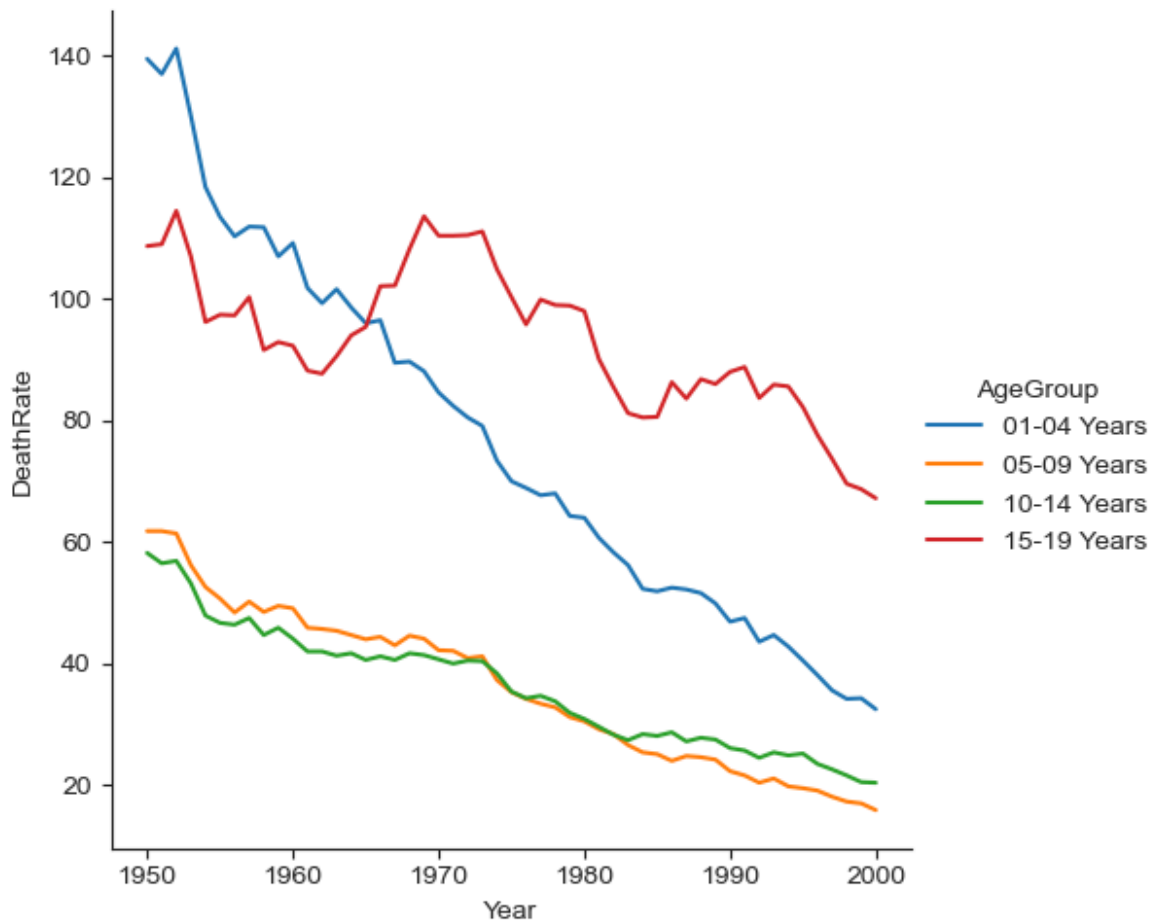
Out[56]: [Text(0.5, 1.0, '15-19 Age Group'), Text(0, 0.5, 'Deaths per 100,000')]



```
In [50]: # Visualize trends in death rates by age group from 1950 to 2000 using a line plot
sns.relplot(data=mortality_data.query('Year >= 1950 and Year <= 2000'), kind='line',
            x='Year', y='DeathRate', hue='AgeGroup')
```

```
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
rt inf values to NaN before operating instead.
  with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning:
use_inf_as_na option is deprecated and will be removed in a future version. Conve
rt inf values to NaN before operating instead.
  with pd.option_context('mode.use_inf_as_na', True):
C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: Th
e figure layout has changed to tight
  self._figure.tight_layout(*args, **kwargs)
```

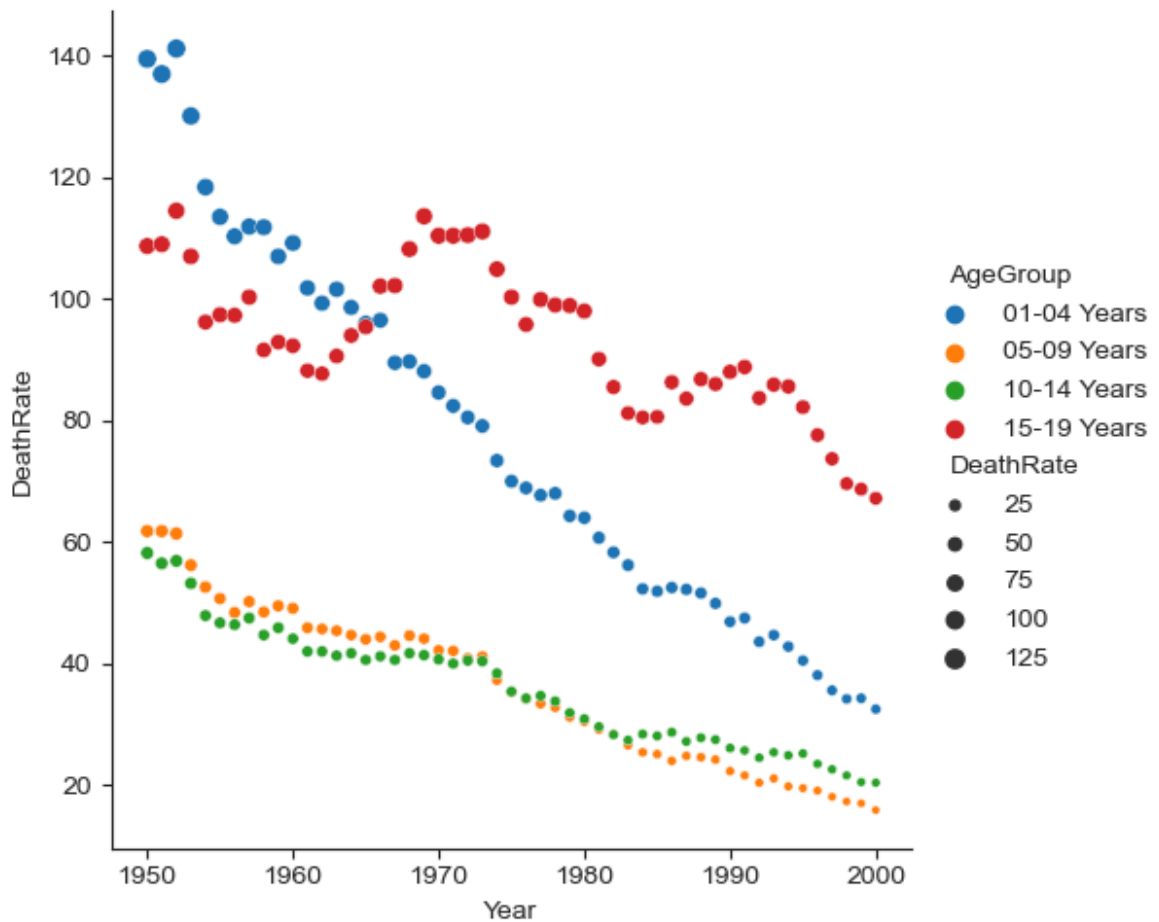
```
Out[50]: <seaborn.axisgrid.FacetGrid at 0x2cdd45357d0>
```



```
In [51]: # Display a scatter plot showing death rates by age group between 1950 and 2000,
sns.relplot(data=mortality_data.query('Year >= 1950 and Year <= 2000'), kind='scatter',
            x='Year', y='DeathRate', hue='AgeGroup', size='DeathRate', sizes=(100, 1000))
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The figure layout has changed to tight
self._figure.tight_layout(*args, **kwargs)

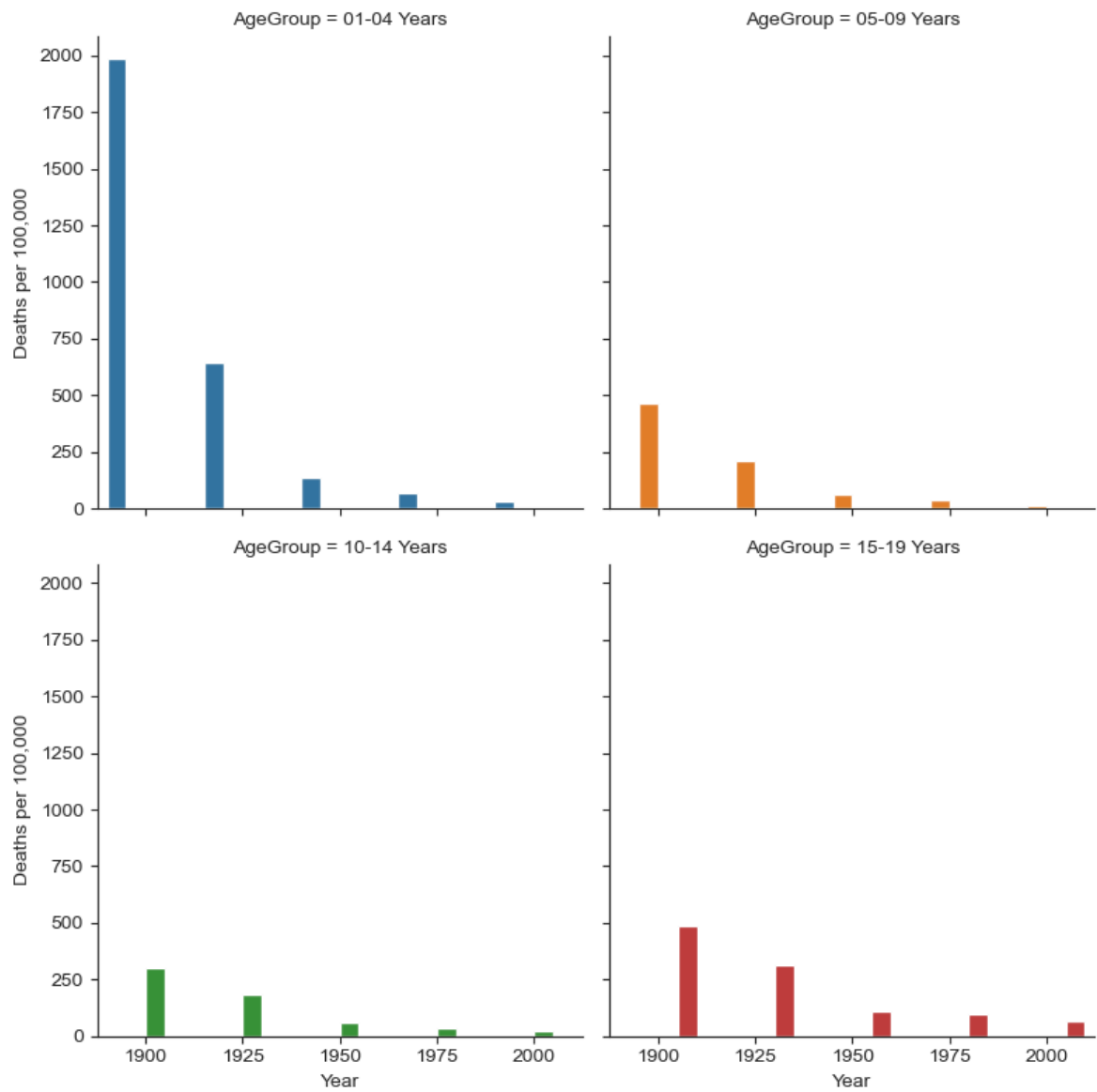
```
Out[51]: <seaborn.axisgrid.FacetGrid at 0x2cdd484d710>
```



```
In [52]: # Generate bar plots for death rates by age group at selected years, saving the
g = sns.catplot(data=mortality_data.query('Year in (1900,1925,1950,1975,2000)'),
               kind='bar', x='Year', y='DeathRate', hue='AgeGroup', legend=False,
               col='AgeGroup', col_wrap=2, height=4)
g.fig.suptitle('Deaths by Age Group', y=1.025)
for ax in g.axes.flat:
    ax.set(ylabel='Deaths per 100,000')
g.savefig('barCharts.png')
```

C:\Users\Dell\ANACONDA\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning: The figure layout has changed to tight
self._figure.tight_layout(*args, **kwargs)

Deaths by Age Group



In []: